



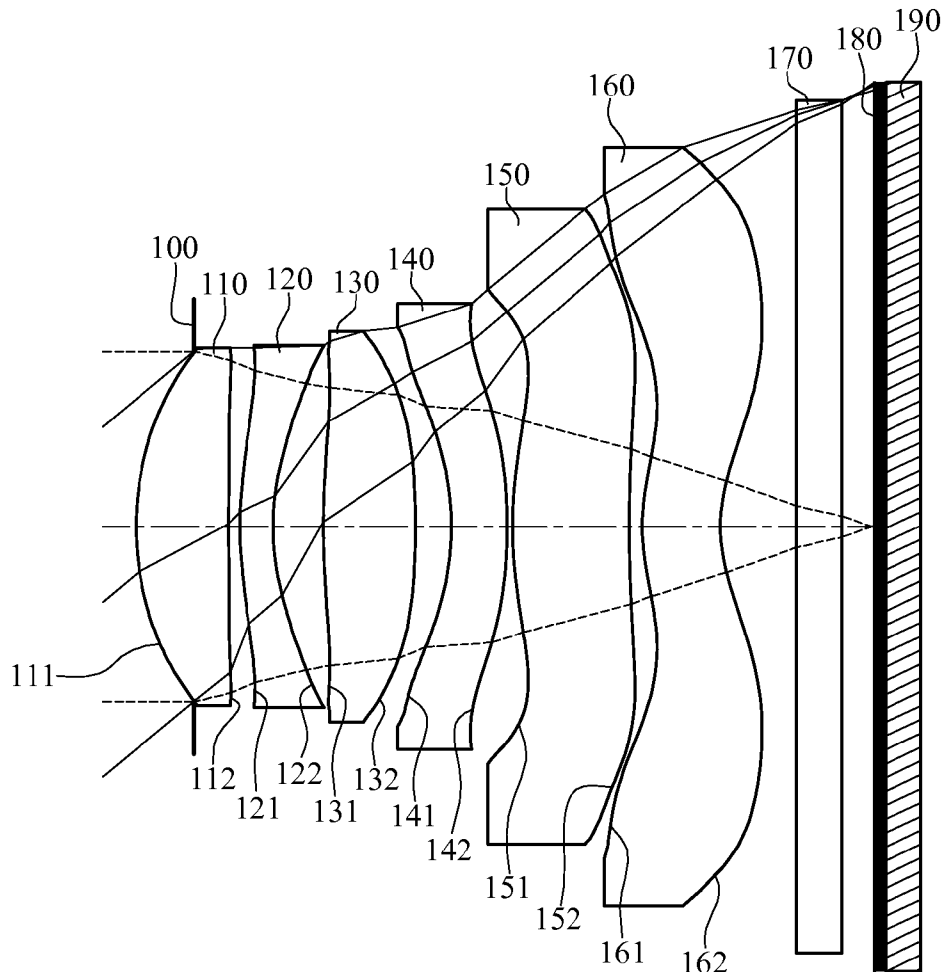
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(19) **United States**(12) **Patent Application Publication**
LIAO et al.(10) **Pub. No.: US 2015/0338607 A1**(43) **Pub. Date: Nov. 26, 2015**(54) **PHOTOGRAPHING OPTICAL LENS
ASSEMBLY, IMAGE CAPTURING UNIT AND
MOBILE DEVICE**(52) **U.S. Cl.**
CPC **G02B 13/0045** (2013.01); **G02B 9/62**
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Tung-Yi HSIEH, Taichung (TW)(21) Appl. No.: **14/337,606**(22) Filed: **Jul. 22, 2014**(30) **Foreign Application Priority Data**

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G02B 9/62 (2006.01)(57) **ABSTRACT**

A photographing optical lens assembly includes, in order from object side to image side, a first lens element, a second lens element, a third lens element, a fourth lens element, a fifth lens element and a sixth lens element. The first lens element has positive refractive power. The second lens element with negative refractive power has an object-side surface being convex in a paraxial region and an image-side surface being concave in a paraxial region. The third and fourth lens elements have refractive power. The fifth lens element with refractive power has an object-side surface being convex in a paraxial region. The sixth lens element with negative refractive power has an image-side surface being concave in a paraxial region, and an object-side surface and the image-side surface of the sixth lens element are aspheric. The image-side surface of the sixth lens element has at least one inflection point.



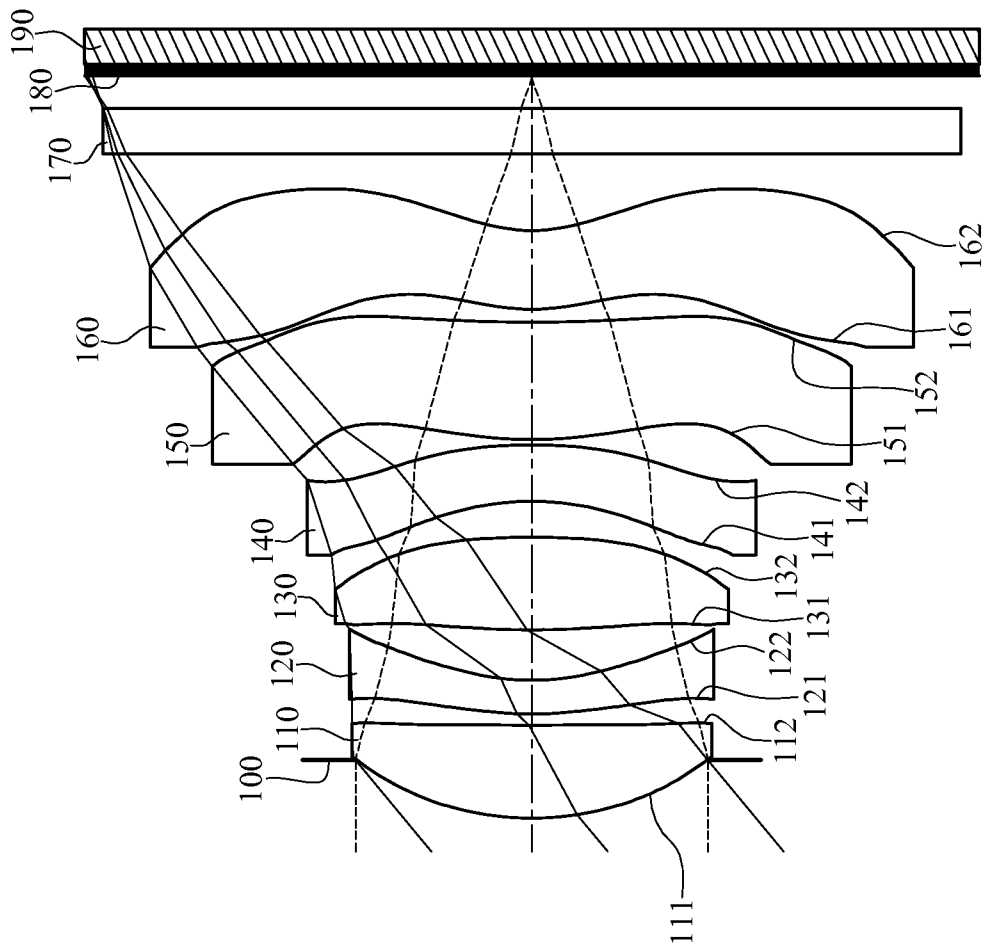


FIG.1

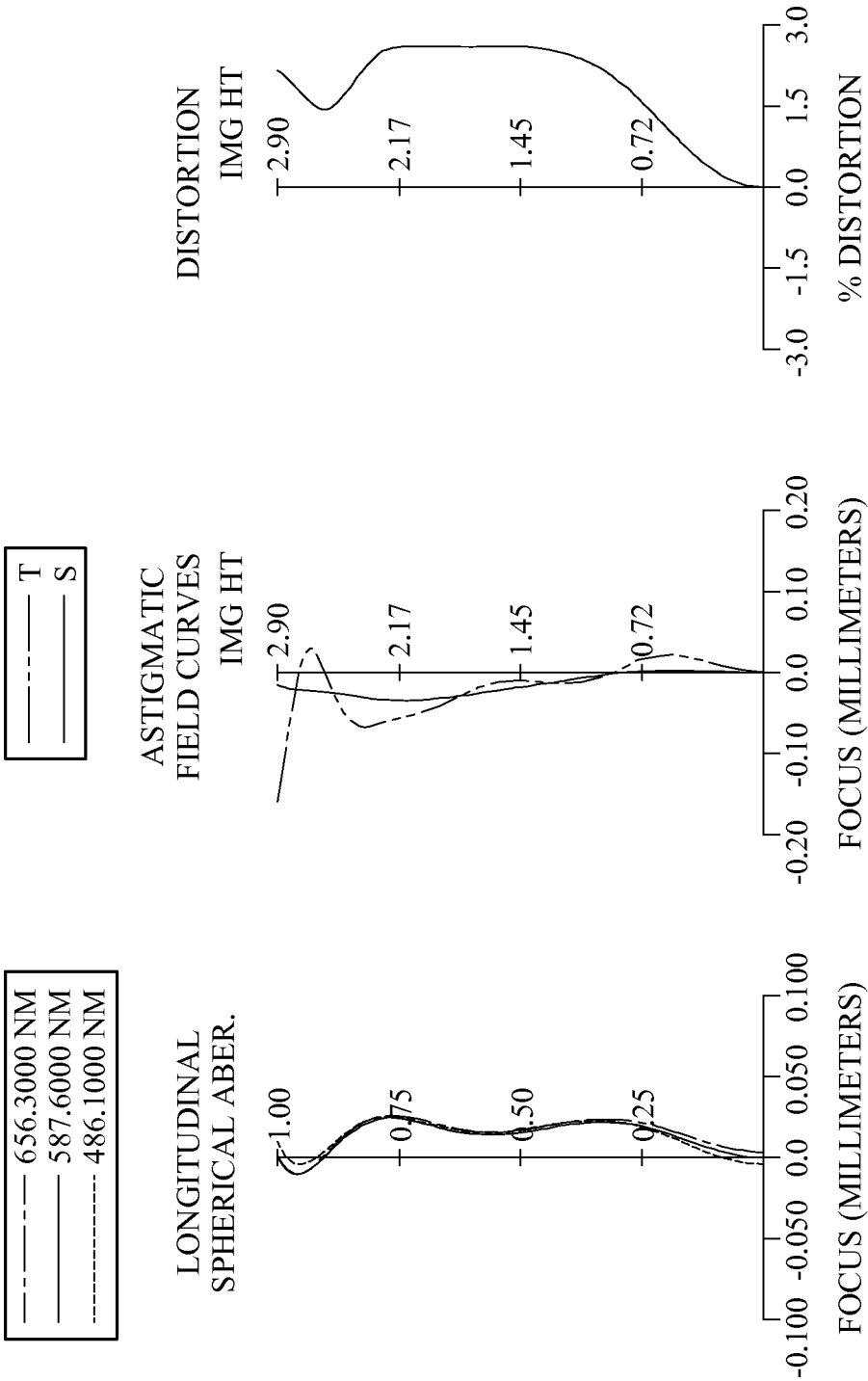


FIG.2

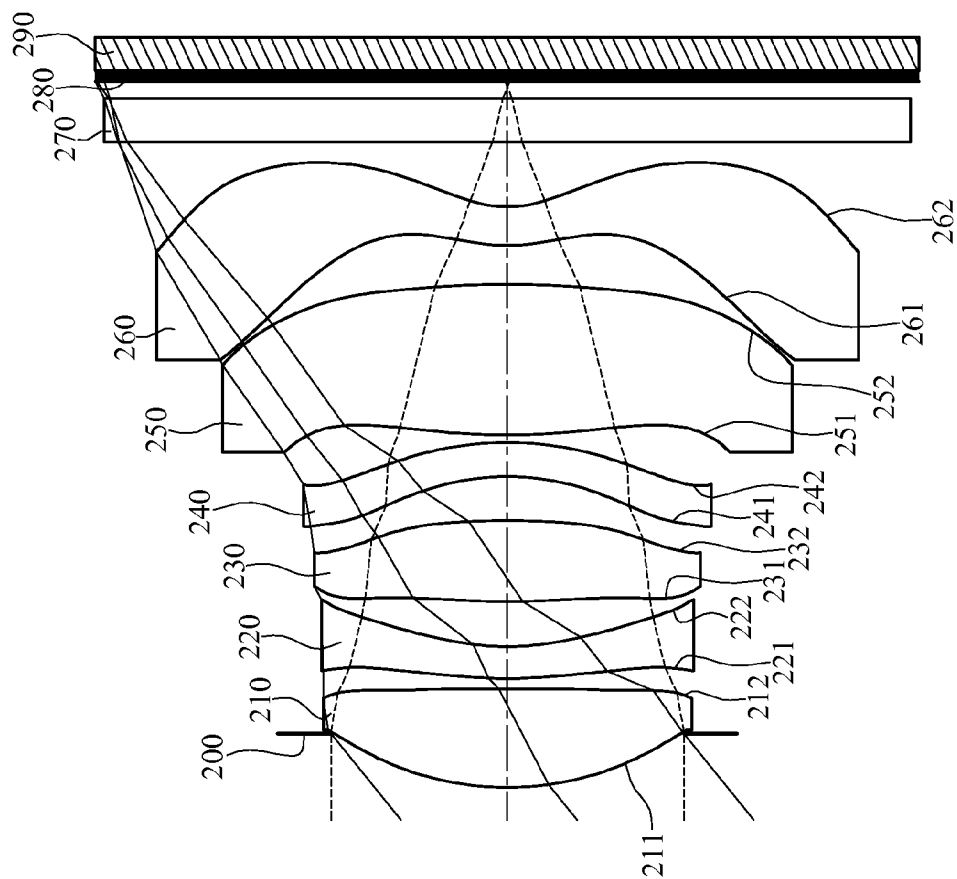


FIG.3

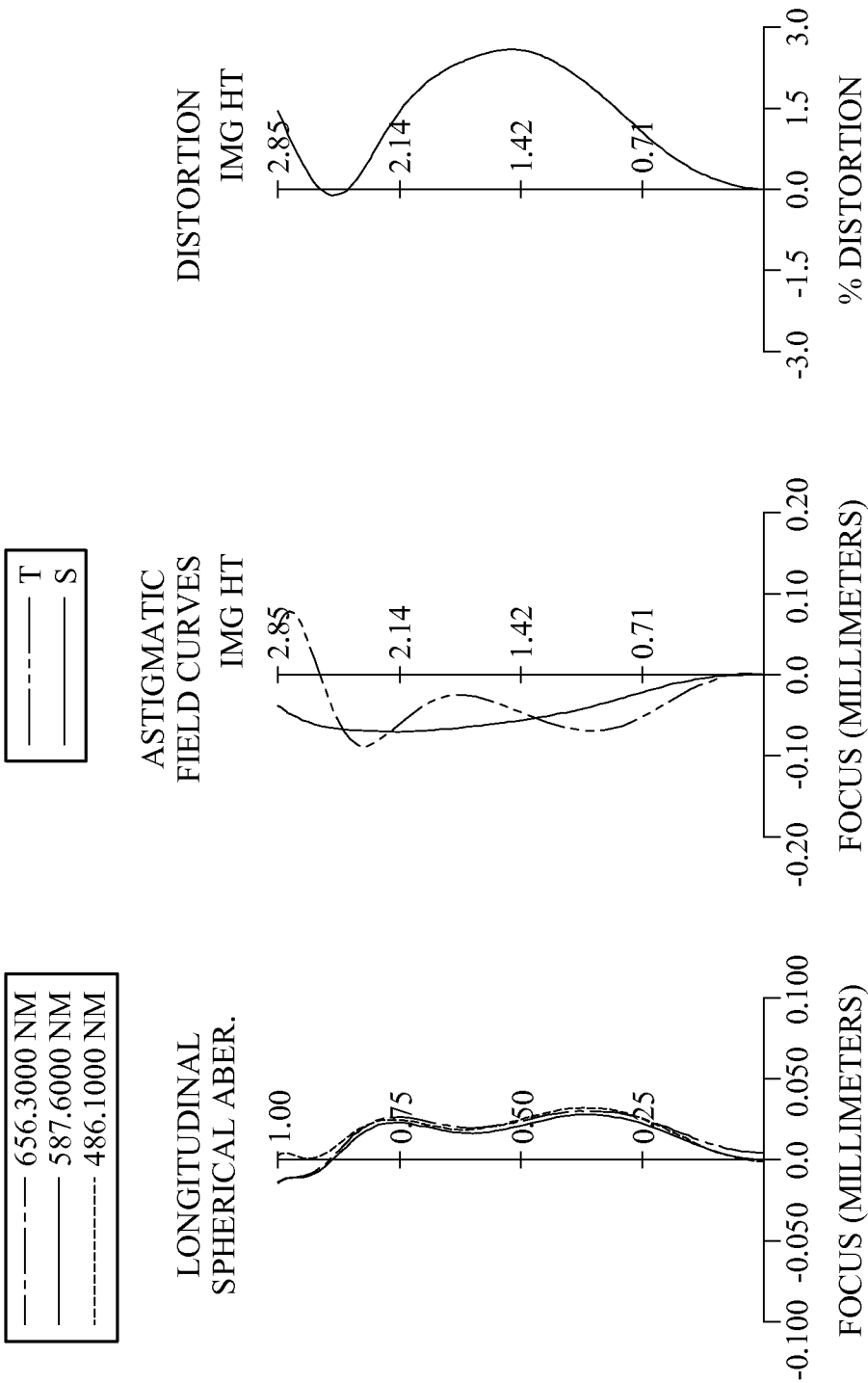


FIG.4

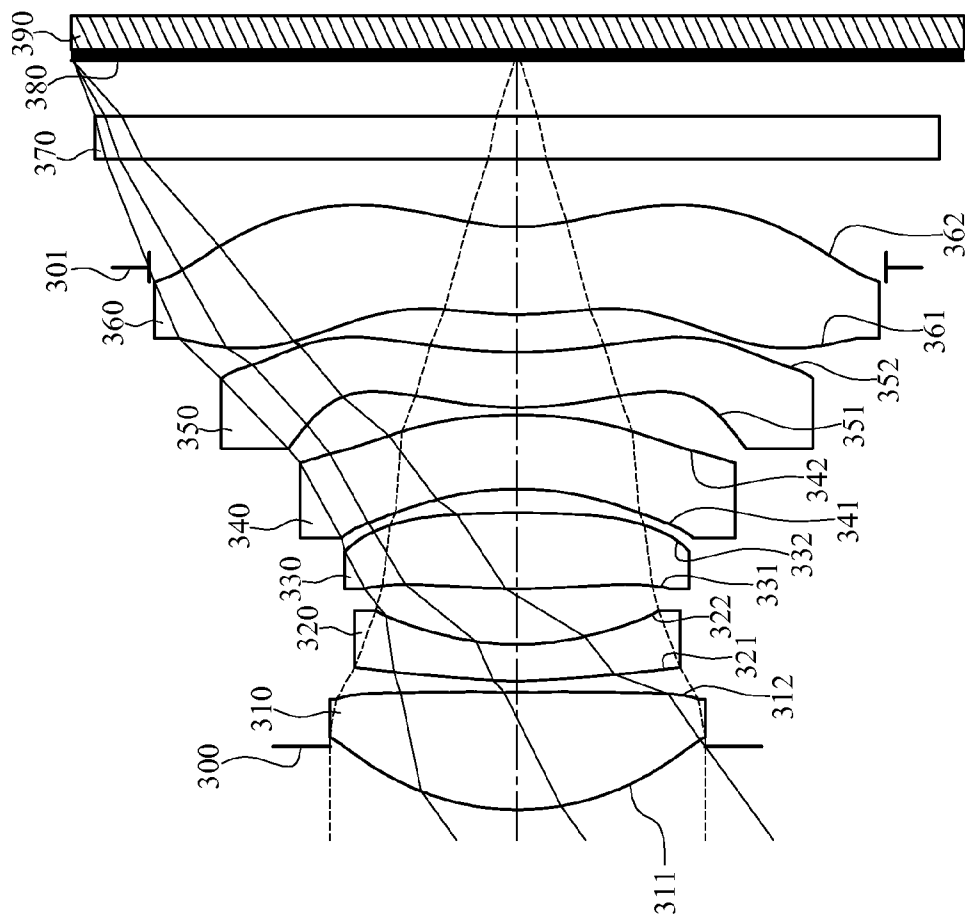


FIG.5

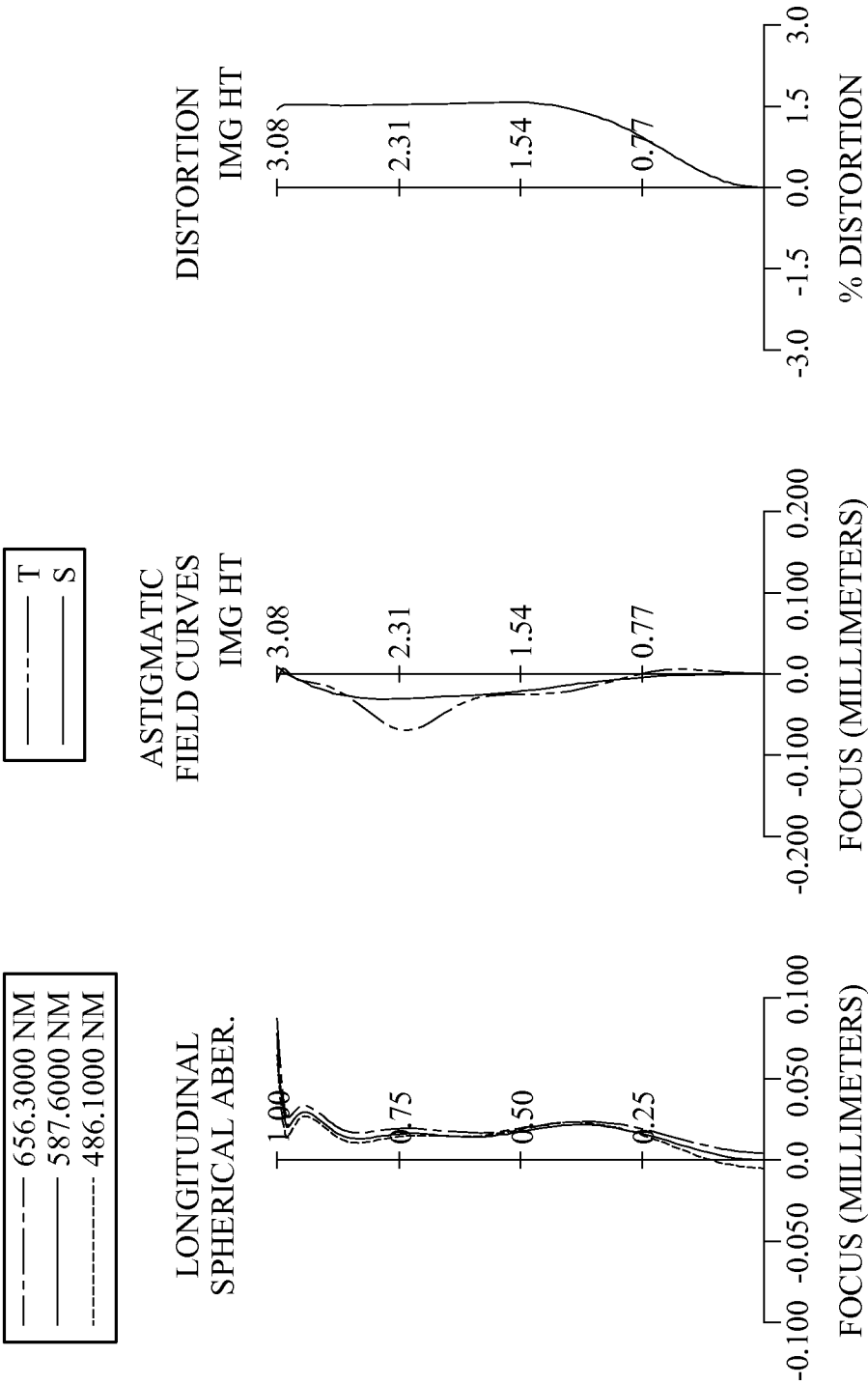


FIG.6

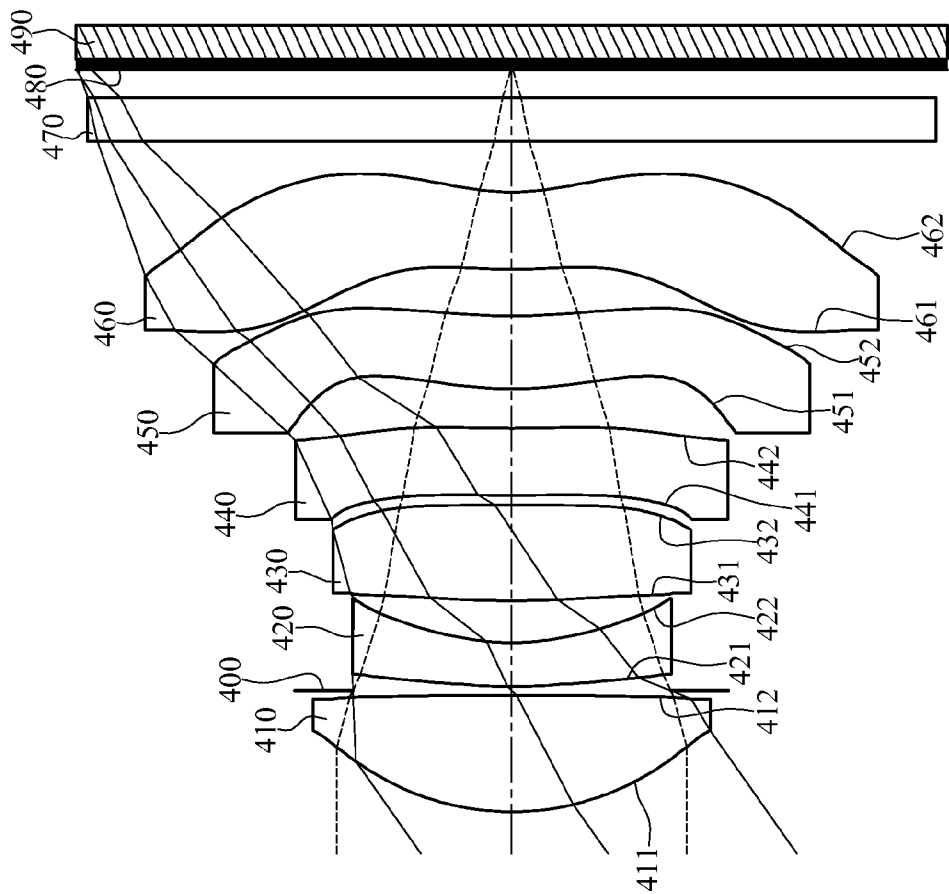
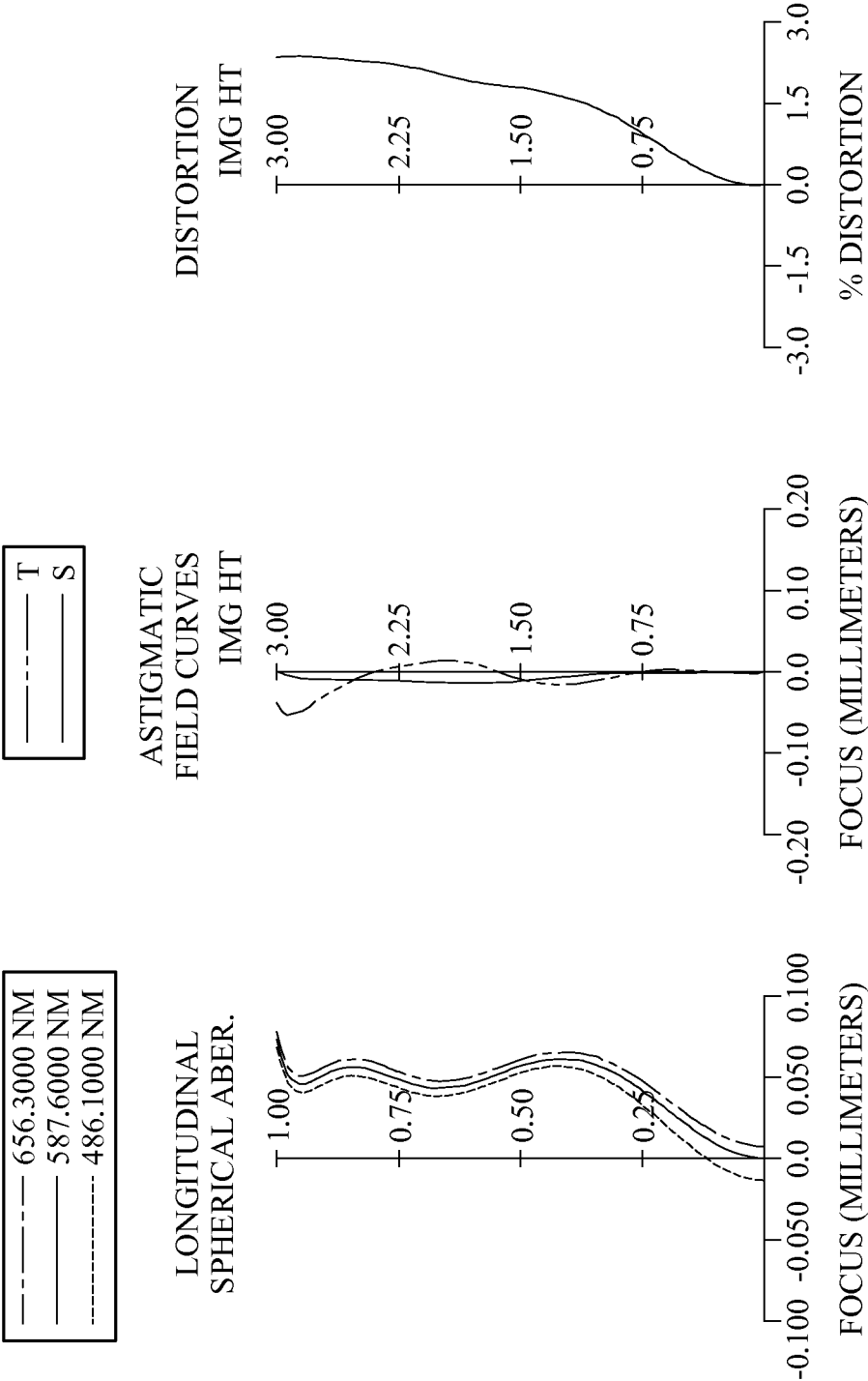


FIG.7



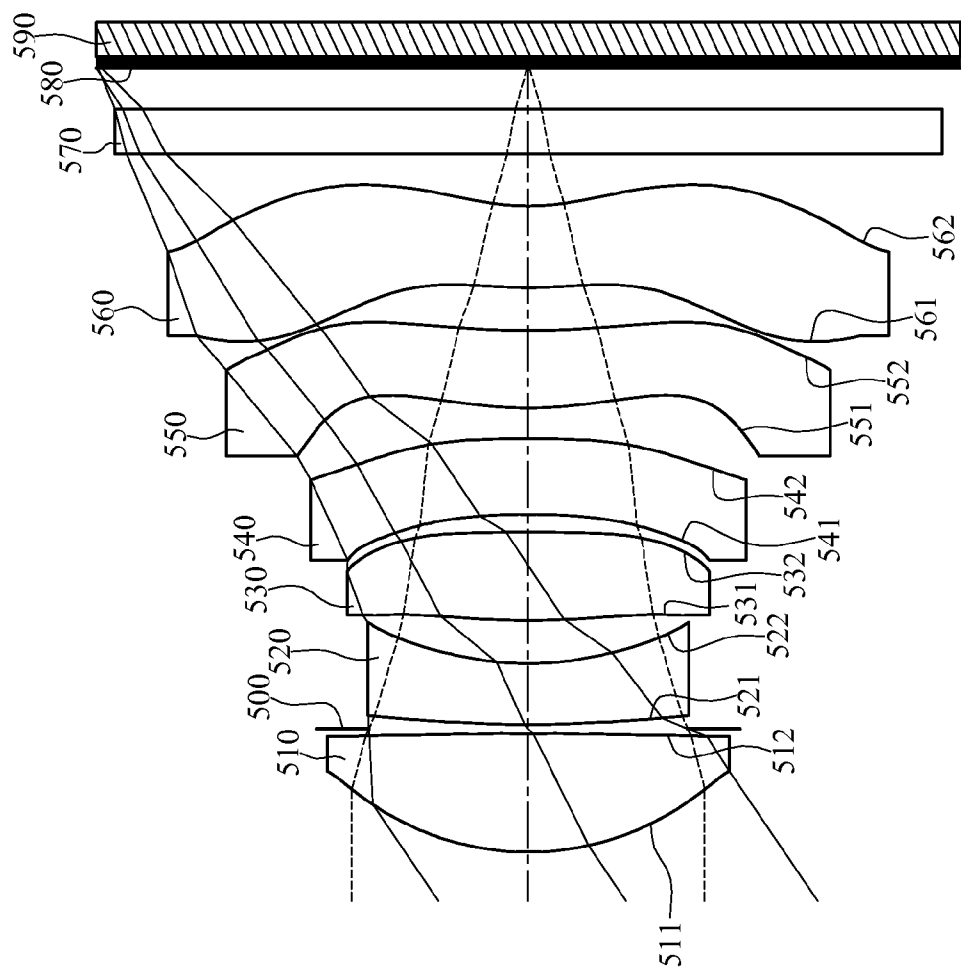


FIG. 9

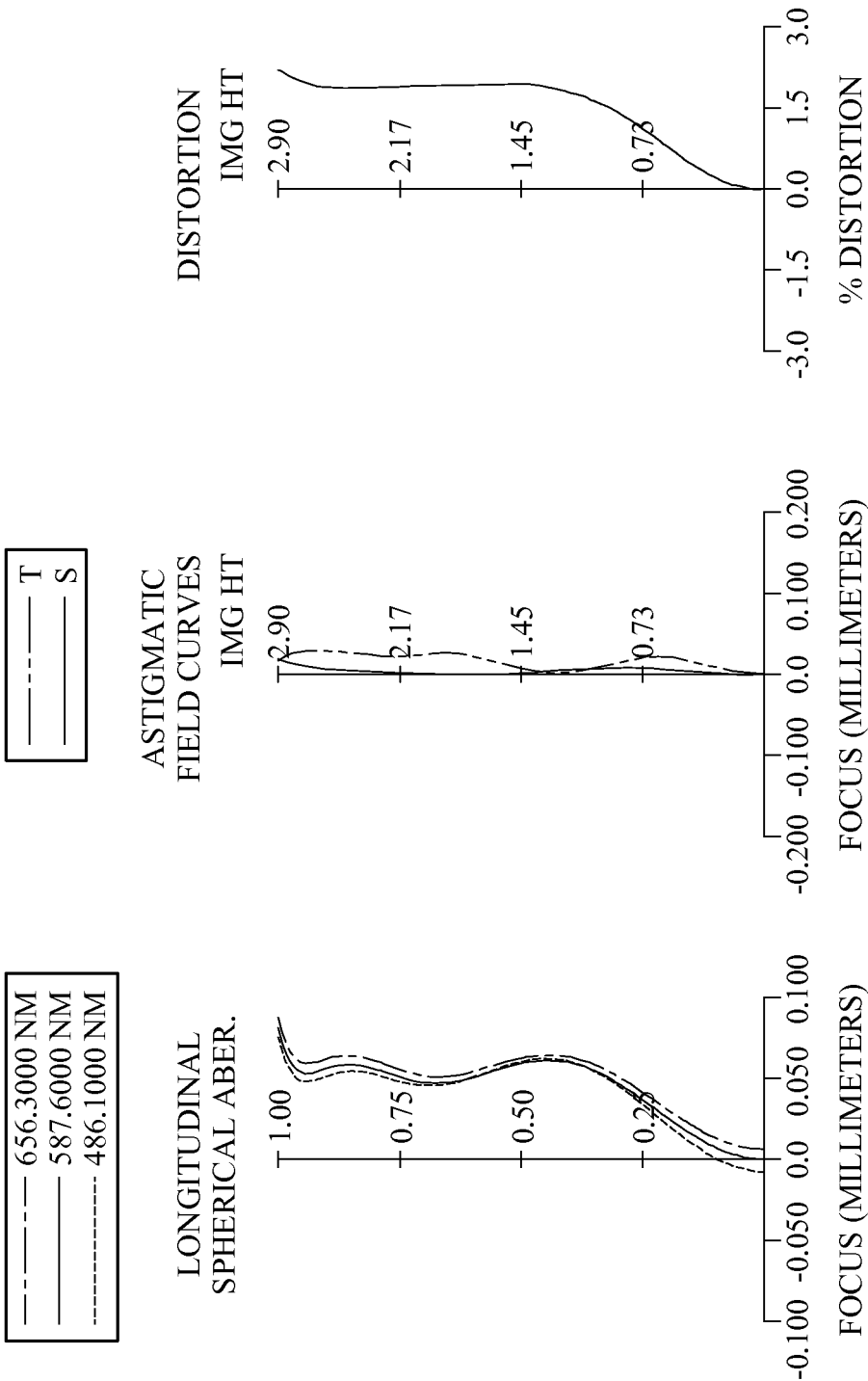


FIG.10

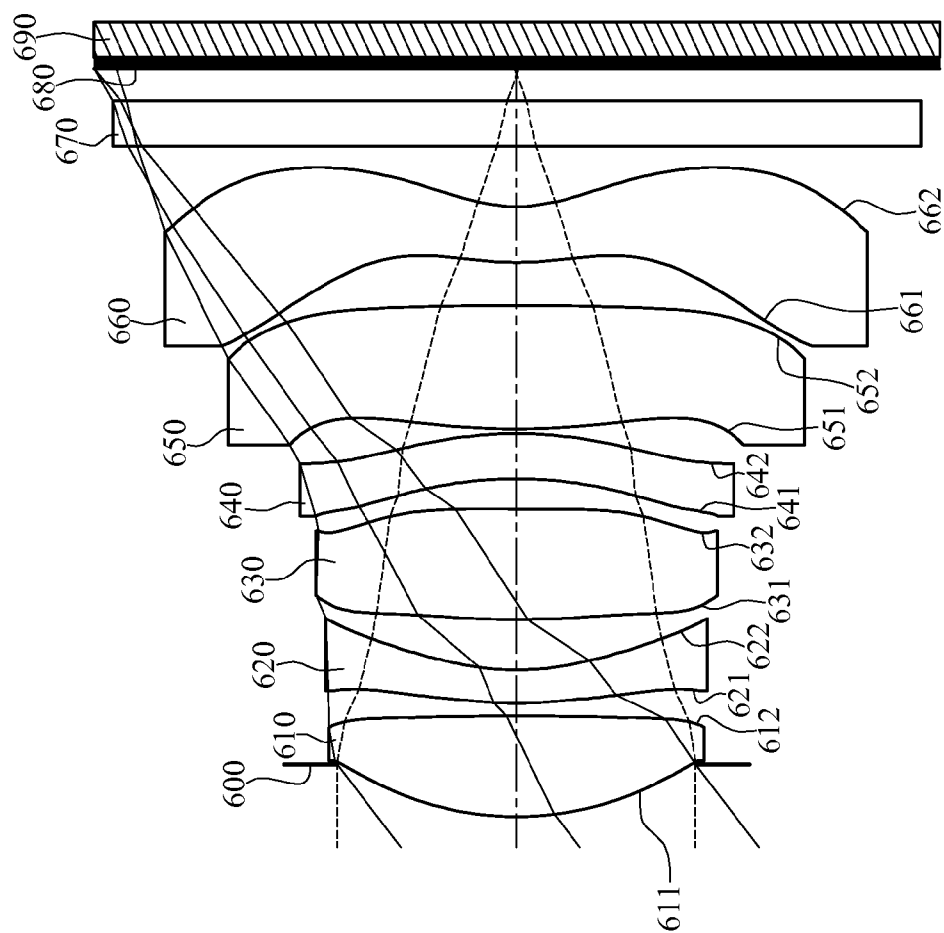
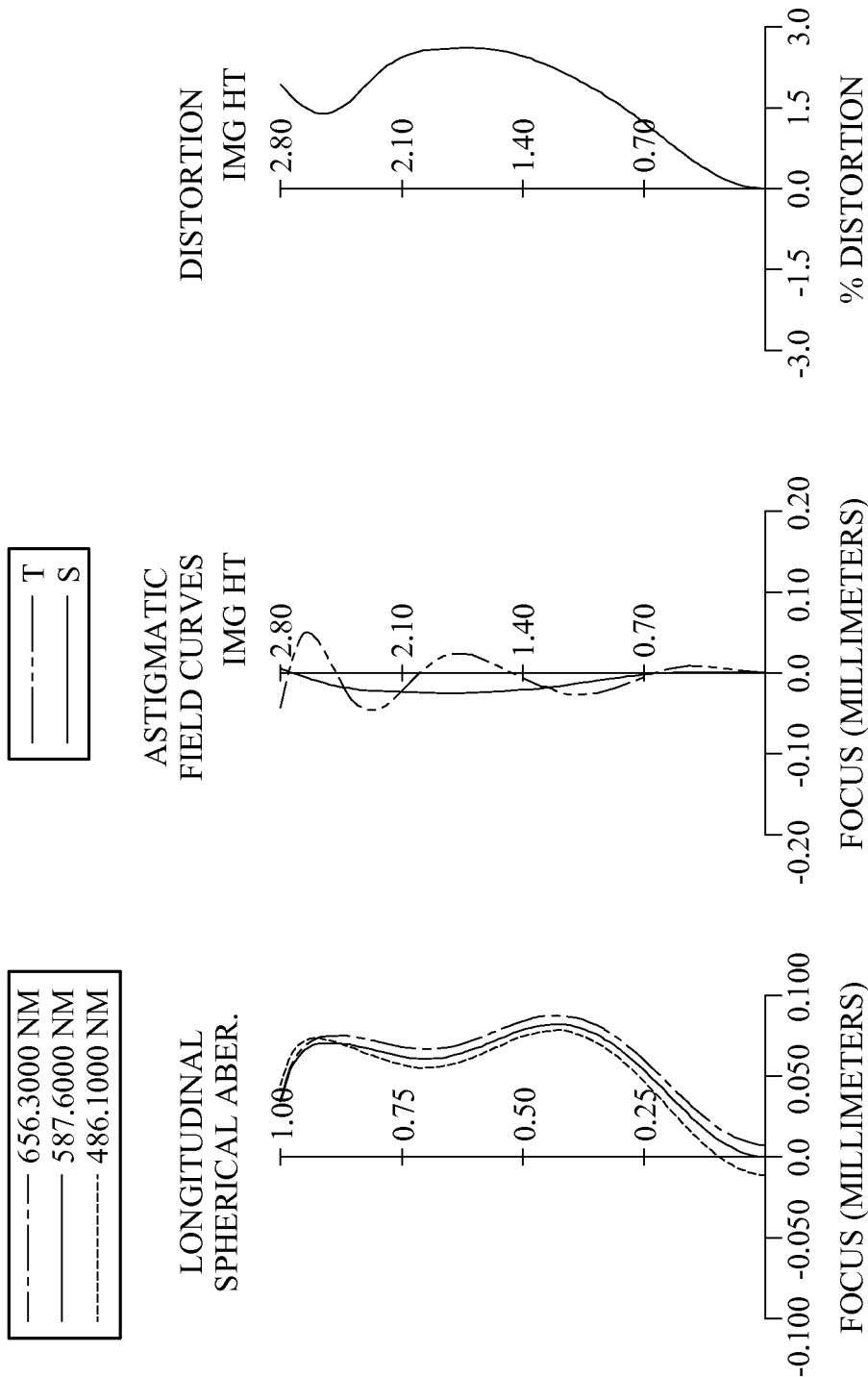


FIG. 11



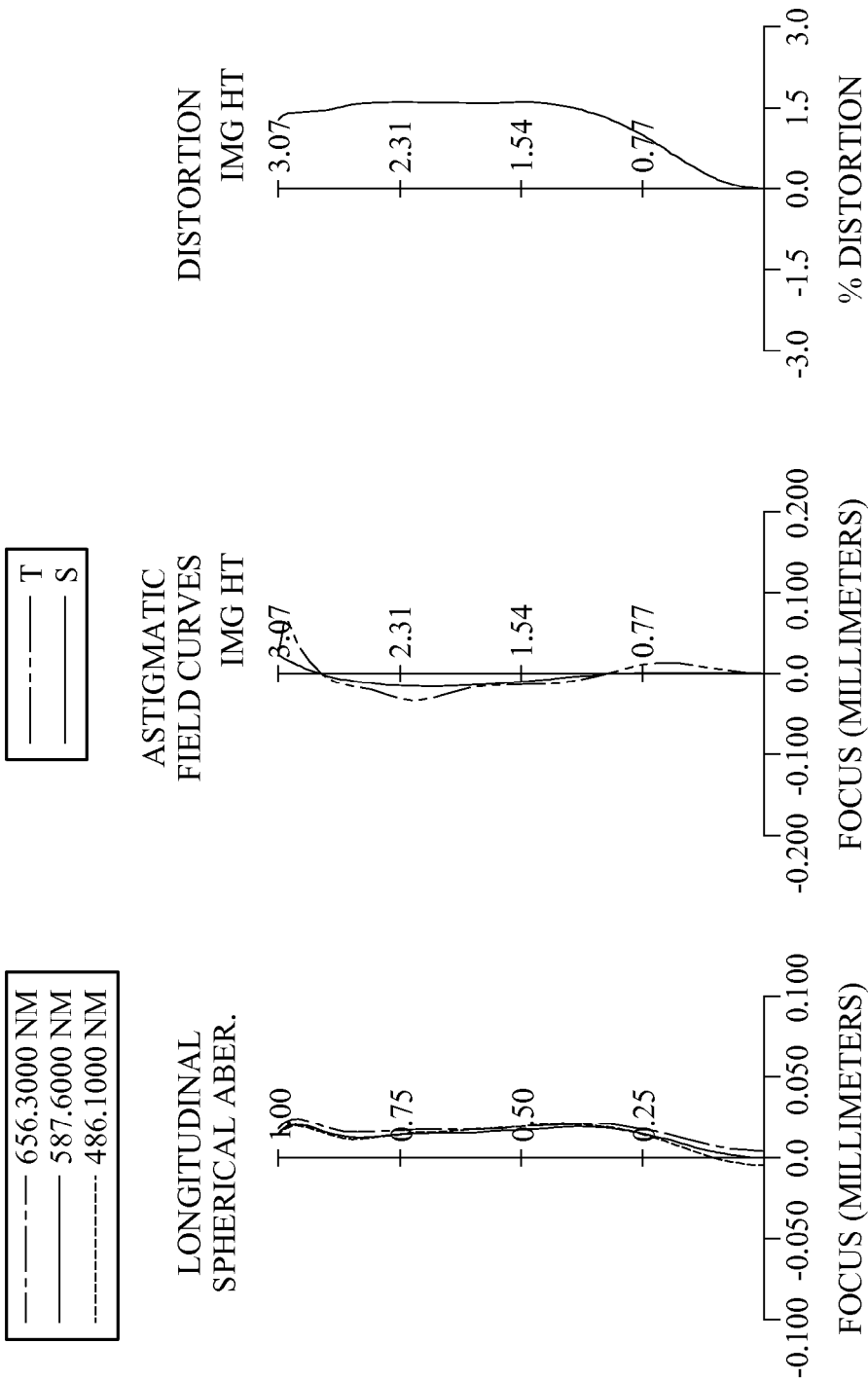


FIG.14

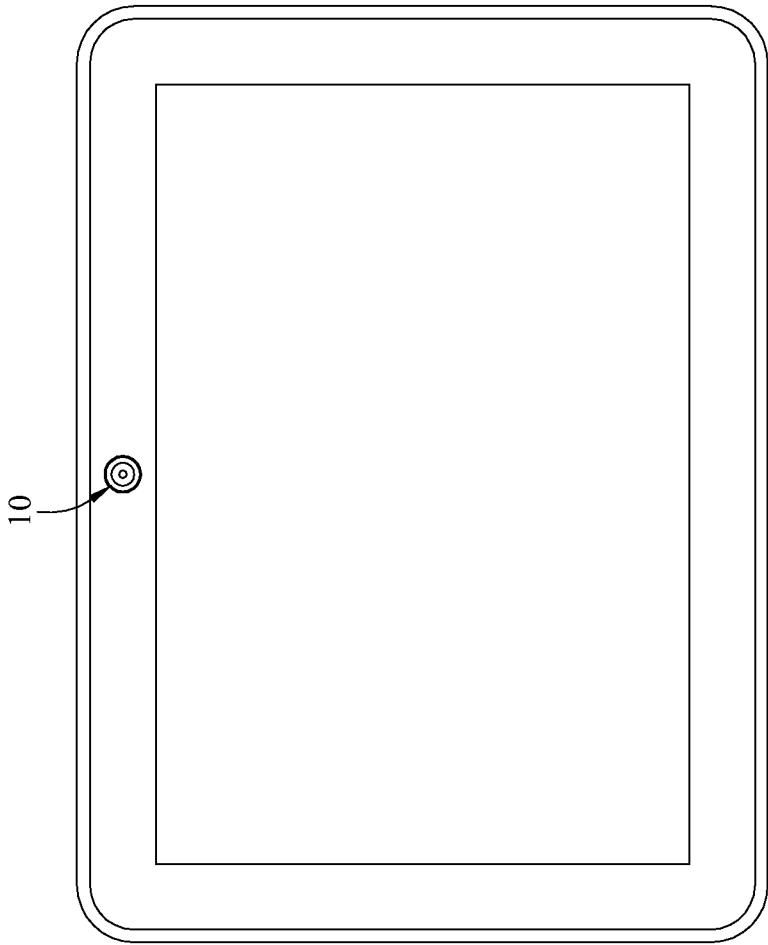


FIG.16

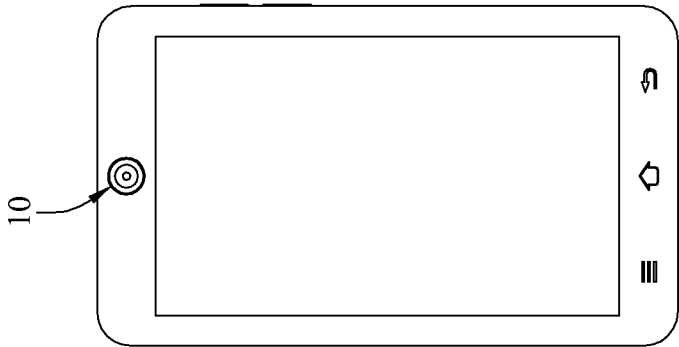


FIG.15

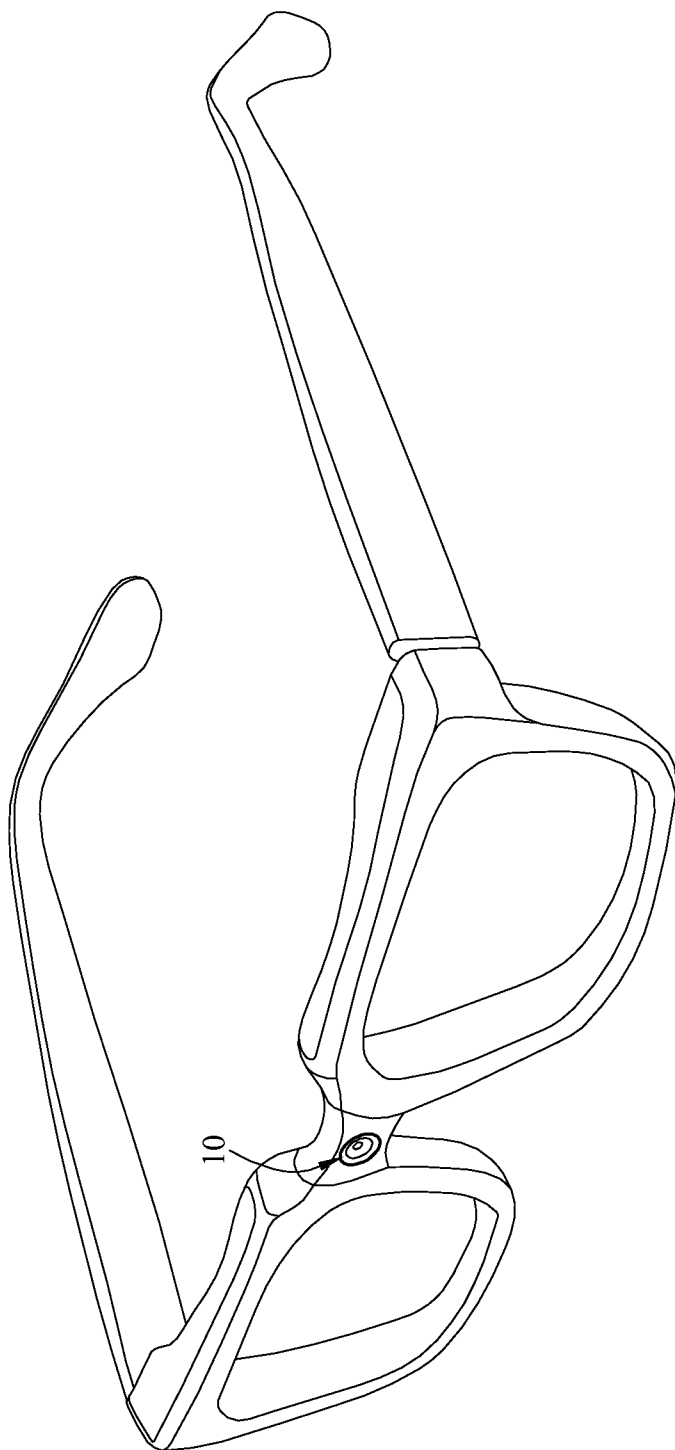


FIG.17

PHOTOGRAPHING OPTICAL LENS ASSEMBLY, IMAGE CAPTURING UNIT AND MOBILE DEVICE

RELATED APPLICATIONS

[0001] This application claims priority to Taiwan Application Serial Number 103118179, filed May 23, 2014, which is incorporated by reference herein in its entirety.

BACKGROUND

[0002] 1. Technical Field

[0003] The present disclosure relates to a photographing optical lens assembly, an image capturing unit and a mobile device, more particularly to a photographing optical lens assembly and an image capturing unit applicable to a mobile device.

[0004] 2. Description of Related Art

[0005] In recent years, with the popularity of mobile terminals having camera functionalities, the demand of miniaturized optical systems has been increasing. The sensor of a conventional optical system is typically a CCD (Charge-Coupled Device) or a CMOS (Complementary Metal-Oxide-Semiconductor) sensor. As the advanced semiconductor manufacturing technologies have allowed the pixel size of sensors to be reduced and compact optical systems have gradually evolved toward the field of higher megapixels, there is an increasing demand for compact optical systems featuring better image quality.

[0006] A conventional optical system employed in a portable electronic product mainly adopts a five-element lens structure. Due to the popularity of mobile terminals with high-end specifications, such as smart phones, tablet personal computers and wearable apparatus, the requirements for high resolution and image quality of present compact optical systems increase significantly. However, the conventional optical systems cannot satisfy these requirements of the compact optical systems.

[0007] Other conventional compact optical systems with six-element lens structure are developed to enhance image quality. However, when the optical system captures an image in a low light condition, the image quality is unclear with low resolution. Furthermore, the interior space of the optical system is utilized inefficiently so that it is unfavorable for keeping the optical system compact.

SUMMARY

[0008] According to one aspect of the present disclosure, a photographing optical lens assembly includes, in order from an object side to an image side, a first lens element, a second lens element, a third lens element, a fourth lens element, a fifth lens element, and a sixth lens element. The first lens element has positive refractive power. The second lens element with negative refractive power has an object-side surface being convex in a paraxial region thereof and an image-side surface being concave in a paraxial region thereof. The third lens element has refractive power. The fourth lens element has refractive power. The fifth lens element with refractive power has an object-side surface being convex in a paraxial region thereof, wherein both of the object-side surface and an image-side surface of the fifth lens element are aspheric. The sixth lens element with negative refractive power has an image-side surface being concave in a paraxial region thereof, wherein the image-side surface of the sixth

lens element has at least one inflection point, both of an object-side surface and the image-side surface of the sixth lens element are aspheric. The photographing optical lens assembly has a total of six lens elements with refractive power. When an entrance pupil diameter of the photographing optical lens assembly is EPD, a sum of the axial distances between any two adjacent lens elements is ΣAT , a curvature radius of the object-side surface of the fifth lens element is $R9$, a curvature radius of the image-side surface of the fifth lens element is $R10$, a curvature radius of the object-side surface of the sixth lens element is $R11$, a curvature radius of the image-side surface of the sixth lens element is $R12$, the following conditions are satisfied:

$$2.20 < EPD/\Sigma AT;$$

$$(R9+R10)/(R9-R10) < -0.15; \text{ and}$$

$$1.20 < (R11+R12)/(R11-R12).$$

[0009] According to another aspect of the present disclosure, an image capturing unit includes the photographing optical lens assembly according to the aforementioned aspect and an image sensor.

[0010] According to still another aspect of the present disclosure, a mobile device includes the image capturing unit according to the aforementioned aspect.

[0011] According to yet another aspect of the present disclosure, a photographing optical lens assembly includes, in order from an object side to an image side, a first lens element, a second lens element, a third lens element, a fourth lens element, a fifth lens element, and a sixth lens element. The first lens element has positive refractive power. The second lens element has negative refractive power. The third lens element has refractive power. The fourth lens element has refractive power. The fifth lens element has refractive power, wherein both of an object-side surface and an image-side surface of the fifth lens element are aspheric. The sixth lens element with refractive power has an image-side surface being concave in a paraxial region thereof, wherein the image-side surface of the sixth lens element has at least one inflection point, both of an object-side surface and the image-side surface of the sixth lens element are aspheric. The photographing optical lens assembly has a total of six lens elements with refractive power. When an entrance pupil diameter of the photographing optical lens assembly is EPD, a sum of the axial distances between any two adjacent lens elements is ΣAT , a curvature radius of the object-side surface of the fifth lens element is $R9$, a curvature radius of the image-side surface of the fifth lens element is $R10$, and the following conditions are satisfied:

$$2.40 < EPD/\Sigma AT; \text{ and}$$

$$(R9+R10)/(R9-R10) < -0.15.$$

[0012] According to still yet another aspect of the present disclosure, a photographing optical lens assembly includes, in order from an object side to an image side, a first lens element, a second lens element, a third lens element, a fourth lens element, a fifth lens element, and a sixth lens element. The first lens element with positive refractive power has an object-side surface being convex in a paraxial region thereof. The second lens element has refractive power. The third lens element has refractive power. The fourth lens element has refractive power. The fifth lens element has refractive power, wherein both of an object-side surface and an image-side

surface of the fifth lens element are aspheric. The sixth lens element with refractive power has an image-side surface being concave in a paraxial region thereof, wherein the image-side surface of the sixth lens element has at least one inflection point, both of an object-side surface and the image-side surface of the sixth lens element are aspheric. The photographing optical lens assembly has a total of six lens elements with refractive power. When an entrance pupil diameter of the photographing optical lens assembly is EPD, a sum of axial distances between any two adjacent lens elements is ΣAT , an axial distance between the object-side surface of the first lens element and an image surface is TL, a maximum image height of the photographing optical lens assembly is ImgH, and the following conditions are satisfied:

$$2.20 < EPD/\Sigma AT;$$

$$0.73 < EPD/ImgH < 1.10; \text{ and}$$

$$TL/ImgH < 2.50.$$

BRIEF DESCRIPTION OF THE DRAWINGS

[0013] The disclosure can be more fully understood by reading the following detailed description of the embodiments, with reference made to the accompanying drawings as follows:

[0014] FIG. 1 is a schematic view of an image capturing unit according to the 1st embodiment of the present disclosure;

[0015] FIG. 2 shows spherical aberration curves, astigmatic field curves and a distortion curve of the image capturing unit according to the 1st embodiment;

[0016] FIG. 3 is a schematic view of an image capturing unit according to the 2nd embodiment of the present disclosure;

[0017] FIG. 4 shows spherical aberration curves, astigmatic field curves and a distortion curve of the image capturing unit according to the 2nd embodiment;

[0018] FIG. 5 is a schematic view of an image capturing unit according to the 3rd embodiment of the present disclosure;

[0019] FIG. 6 shows spherical aberration curves, astigmatic field curves and a distortion curve of the image capturing unit according to the 3rd embodiment;

[0020] FIG. 7 is a schematic view of an image capturing unit according to the 4th embodiment of the present disclosure;

[0021] FIG. 8 shows spherical aberration curves, astigmatic field curves and a distortion curve of the image capturing unit according to the 4th embodiment;

[0022] FIG. 9 is a schematic view of an image capturing unit according to the 5th embodiment of the present disclosure;

[0023] FIG. 10 shows spherical aberration curves, astigmatic field curves and a distortion curve of the image capturing unit according to the 5th embodiment;

[0024] FIG. 11 is a schematic view of an image capturing unit according to the 6th embodiment of the present disclosure;

[0025] FIG. 12 shows spherical aberration curves, astigmatic field curves and a distortion curve of the image capturing unit according to the 6th embodiment;

[0026] FIG. 13 is a schematic view of an image capturing unit according to the 7th embodiment of the present disclosure;

[0027] FIG. 14 shows spherical aberration curves, astigmatic field curves and a distortion curve of the image capturing unit according to the 7th embodiment;

[0028] FIG. 15 shows a mobile device according to an embodiment;

[0029] FIG. 16 shows a mobile device according to an embodiment; and

[0030] FIG. 17 shows a mobile device according to an embodiment.

DETAILED DESCRIPTION

[0031] An photographing optical lens assembly includes, in order from an object side to an image side, a first lens element, a second lens element, a third lens element, a fourth lens element, a fifth lens element, and a sixth lens element. The photographing optical lens assembly has a total of six lens elements with refractive power.

[0032] The first lens element with positive refractive power can have an object-side surface being convex in a paraxial region thereof. Therefore, it provides the ability to reduce the total track length thereof so as to keep the photographing optical lens assembly compact.

[0033] The second lens element can have negative refractive power. The second lens element can have an object-side surface being convex in a paraxial region thereof and an image-side surface being concave in a paraxial region thereof. Therefore, it is favorable for correcting the astigmatism from the first lens element so as to enhance the image quality.

[0034] The third lens element can have positive refractive power. The third lens element can have an object-side surface being convex in a paraxial region thereof. Therefore, it is favorable for reducing the sensitivity of the photographing optical lens assembly and correcting the spherical aberration so as to further improve the image quality.

[0035] The fourth lens element can have negative refractive power. The fourth lens element can have an object-side surface being concave in a paraxial region thereof and an image-side surface being convex in a paraxial region thereof. Therefore, it is favorable for effectively correcting the Petzval's Sum of the photographing optical lens assembly so that the image surface can be more flat. Furthermore, it is favorable for correcting the aberration.

[0036] The fifth lens element can have positive refractive power. The fifth lens element can have an object-side surface being convex in a paraxial region thereof and an image-side surface being concave in paraxial region thereof. The object-side surface of the fifth lens element can have at least one inflection point. Therefore, it is favorable for preventing the spherical aberration from becoming too large and reducing the astigmatism. Furthermore, it is also favorable for correcting the off-axis aberration.

[0037] The sixth lens element can have negative refractive power. The sixth lens element can have an object-side surface being convex in a paraxial region thereof and an image-side surface being concave in a paraxial region thereof. The object-side surface of the sixth lens element can have at least one concave shape in an off-axis region thereof. The image-side surface of the sixth lens element has at least one inflection point. Therefore, it is favorable for the principal point of the photographing optical lens assembly being positioned away from the image side and reducing a back focal length of

the photographing optical lens assembly so as to keep a compact size thereof. Furthermore, it is favorable for effectively reducing the incident angle of the light projecting onto the image sensor, so that it is favorable for improving the responding sensitivity of the image sensor and further correcting the off-axis aberration.

[0038] When an entrance pupil diameter of the photographing optical lens assembly is EPD, a sum of the axial distances between any two adjacent lens elements is ΣAT (that is, the sum of an axial distance between the first lens element and the second lens element, an axial distance between the second lens element and the third lens element, an axial distance between the third lens element and the fourth lens element, an axial distance between the fourth lens element and the fifth lens element, and an axial distance between the fifth lens element and the sixth lens element), the following condition is satisfied: $2.20 < EPD/\Sigma AT$. Therefore, the lens elements can be tightly arranged so that it is favorable for reducing the total track length thereof and providing sufficient amount of incident light, thereby increasing the relative illumination of the image. Preferably, the following condition is satisfied: $2.40 < EPD/\Sigma AT$.

[0039] When a curvature radius of the object-side surface of the fifth lens element is R9, a curvature radius of the image-side surface of the fifth lens element is R10, the following condition is satisfied: $(R9+R10)/(R9-R10) < -0.15$. Therefore, it is favorable for correcting the spherical aberration and the astigmatism so as to enhance the image quality. Preferably, the following condition is satisfied: $(R9+R10)/(R9-R10) \leq -1.00$.

[0040] When a curvature radius of the object-side surface of the sixth lens element is R11, a curvature radius of the image-side surface of the sixth lens element is R12, the following condition is satisfied: $1.20 < (R11+R12)/(R11-R12)$. Therefore, it is favorable for correcting the astigmatism so as to enhance the image quality.

[0041] When the entrance pupil diameter of the photographing optical lens assembly is EPD, a maximum image height of the photographing optical lens assembly (half of a diagonal length of an effective photosensitive area of an image sensor) is ImgH, the following condition is satisfied: $0.65 < EPD/ImgH < 1.50$. Therefore, it is favorable to providing sufficient amount of incident light to enhance the resolution. Preferably, the following condition is satisfied: $0.78 < EPD/ImgH < 0.90$.

[0042] When an axial distance between the object-side surface of the first lens element and an image surface is TL, the maximum image height of the photographing optical lens assembly is ImgH, the following condition is satisfied: $TL/ImgH < 2.50$. Therefore, it is favorable for keeping the photographing optical lens assembly compact so as to be equipped on portable devices.

[0043] When an f-number of the photographing optical lens assembly is Fno, the following condition is satisfied: $Fno < 1.85$. Therefore, it is favorable for obtaining a proper aperture for receiving sufficient incoming light, thereby increasing the image quality while the photographing optical lens assembly is in a low light condition with a shutter in high speed. Preferably, the following condition is satisfied: $Fno < 1.65$. More preferably, the following condition is satisfied: $Fno < 1.55$.

[0044] When a focal length of the second lens element is f2, a focal length of the third lens element is f3, the following conditions are satisfied: $-1.50 < f3/f2$. Therefore, it is favor-

able for balancing the refractive power of the photographing optical lens assembly to enhance the resolution.

[0045] When an Abbe number of the second lens element is V2, an Abbe number of the third lens element is V3, an Abbe number of the fourth lens element is V4, the following condition is satisfied: $(V2+V4)/V3 < 1.00$. Therefore, it is favorable for correcting the chromatic aberration.

[0046] When a curvature radius of the object-side surface of the fourth lens element is R7, a curvature radius of the image-side surface of the fourth lens element is R8, the following condition is satisfied: $|(R7-R8)/(R7+R8)| < 0.50$. Therefore, it is favorable for correcting the aberration so as to enhance the image quality.

[0047] When the axial distance between the object-side surface of the first lens element and the image surface is TL, the following condition is satisfied: $TL < 6.5$ mm. Therefore, it is favorable for effectively reducing the total track length thereof so as to equip the photographing optical lens assembly on portable devices.

[0048] There is an air gap in a paraxial region between any two of the first lens element, the second lens element, the third lens element, the fourth lens element, the fifth lens element and the sixth lens element that are adjacent to each other. Therefore, it is favorable for effectively assembling the lens elements without interference with each other and reducing the assembly difficulties.

[0049] According to the photographing optical lens assembly of the present disclosure, an image surface of the photographing optical lens assembly, based on the corresponding image sensor, can be flat or curved especially a curved surface being concave facing towards the object side.

[0050] According to the photographing optical lens assembly of the present disclosure, the lens elements thereof can be made of glass or plastic material. When the lens elements are made of glass material, the distribution of the refractive power of the photographing optical lens assembly may be more flexible to design. When the lens elements are made of plastic material, the manufacturing cost can be effectively reduced. Furthermore, surfaces of each lens element can be arranged to be aspheric, since the aspheric surface of the lens element is easy to form a shape other than spherical surface so as to have more controllable variables for eliminating the aberration thereof, and to further decrease the required number of the lens elements. Therefore, the total track length of the photographing optical lens assembly can also be reduced.

[0051] According to the photographing optical lens assembly of the present disclosure, each of an object-side surface and an image-side surface has a paraxial region and an off-axis region. The paraxial region refers to the region of the surface where light rays travel close to the optical axis, and the off-axis region refers to the region of the surface away from the paraxial region. Particularly, when the lens element has a convex surface, it indicates that the surface is convex in the paraxial region thereof; when the lens element has a concave surface, it indicates that the surface is concave in the paraxial region thereof.

[0052] According to the photographing optical lens assembly of the present disclosure, the photographing optical lens assembly can include at least one stop, such as an aperture stop, a glare stop or a field stop. Said glare stop or said field stop is set for eliminating the stray light and thereby improving the image quality thereof.

[0053] According to the photographing optical lens assembly of the present disclosure, an aperture stop can be config-

ured as a front stop or a middle stop. A front stop disposed between an imaged object and the first lens element can provide a longer distance between an exit pupil of the photographing optical lens assembly and the image surface and thereby improves the image-sensing efficiency of an image sensor. A middle stop disposed between the first lens element and the image surface is favorable for enlarging the field of view of the photographing optical lens assembly and thereby provides a wider field of view for the same.

[0054] According to the present disclosure, an image capturing unit is provided. The image capturing unit includes the photographing optical lens assembly according to the aforementioned photographing optical lens assembly of the present disclosure, and an image sensor, wherein the image sensor is disposed on the image side of the aforementioned photographing optical lens assembly, that is, the image sensor can be disposed on or near an image surface of the aforementioned photographing optical lens assembly. In some embodiments, the image capturing unit can further include a barrel member, a holding member or a combination thereof.

[0055] In FIG. 15, FIG. 16 and FIG. 17, an image capturing device 10 may be installed in, but not limited to, a mobile terminal, including a smart phone (FIG. 15), a tablet personal computer (FIG. 16) or a wearable device (FIG. 17). The three exemplary figures of different kinds of mobile terminal are only exemplary figures for showing the image capturing device of present disclosure installing in a mobile terminal and is not limited thereto. In some embodiments, the mobile terminal can further include, but not limited to, a display, a control unit, a random access memory unit (RAM), a read only memory unit (ROM) or a combination thereof.

[0056] According to the photographing optical lens assembly of the present disclosure, the photographing optical lens assembly can be optionally applied to moving focus optical systems. Furthermore, the photographing optical lens assembly is featured with good capability in aberration corrections and high image quality, and can be applied to 3D (three-dimensional) image capturing applications, in products such as digital cameras, mobile devices, digital tablets, wearable devices and other mobile terminals. According to the above description of the present disclosure, the following specific embodiments are provided for further explanation.

1st Embodiment

[0057] FIG. 1 is a schematic view of an image capturing unit according to the 1st embodiment of the present disclosure. FIG. 2 shows, in order from left to right, spherical aberration curves, astigmatic field curves and a distortion curve of the image capturing unit according to the 1st embodiment. In FIG. 1, the image capturing unit includes the photographing optical lens assembly (its reference numeral is omitted) of the present disclosure and an image sensor 190. The photographing optical lens assembly includes, in order from an object side to an image side, an aperture stop 100, a first lens element 110, a second lens element 120, a third lens element 130, a fourth lens element 140, a fifth lens element 150, a sixth lens element 160, an IR-cut filter 170 and an image surface 180, wherein the photographing optical lens assembly has a total of six lens elements (110-160) with refractive power.

[0058] The first lens element 110 with positive refractive power has an object-side surface 111 being convex in a paraxial region thereof and an image-side surface 112 being concave in a paraxial region thereof. The first lens element

110 is made of plastic material and has the object-side surface 111 and the image-side surface 112 being both aspheric.

[0059] The second lens element 120 with negative refractive power has an object-side surface 121 being convex in a paraxial region thereof and an image-side surface 122 being concave in a paraxial region thereof. The second lens element 120 is made of plastic material and has the object-side surface 121 and the image-side surface 122 being both aspheric.

[0060] The third lens element 130 with positive refractive power has an object-side surface 131 being convex in a paraxial region thereof and an image-side surface 132 being convex in a paraxial region thereof. The third lens element 130 is made of plastic material and has the object-side surface 131 and the image-side surface 132 being both aspheric.

[0061] The fourth lens element 140 with negative refractive power has an object-side surface 141 being concave in a paraxial region thereof and an image-side surface 142 being convex in a paraxial region thereof. The fourth lens element 140 is made of plastic material and has the object-side surface 141 and the image-side surface 142 being both aspheric.

[0062] The fifth lens element 150 with positive refractive power has an object-side surface 151 being convex in a paraxial region thereof and an image-side surface 152 being concave in a paraxial region thereof. The object-side surface 151 has at least one inflection point. The fifth lens element 150 is made of plastic material and has the object-side surface 151 and the image-side surface 152 being both aspheric.

[0063] The sixth lens element 160 with negative refractive power has an object-side surface 161 being convex in a paraxial region thereof and an image-side surface 162 being concave in a paraxial region thereof. The object-side surface 161 has at least one concave shape in an off-axis region thereof. The image-side surface 162 has at least one inflection point. The sixth lens element 160 is made of plastic material and has the object-side surface 161 and the image-side surface 162 being both aspheric.

[0064] The IR-cut filter 170 is made of glass and located between the sixth lens element 160 and the image surface 180, and will not affect the focal length of the photographing optical lens assembly. The image sensor 190 is disposed on or near the image surface 180 of the photographing optical lens assembly.

[0065] The equation of the aspheric surface profiles of the aforementioned lens elements of the 1st embodiment is expressed as follows:

$$X(Y) = (Y^2 / R) / (1 + \sqrt{1 - (1 + k) \times (Y / R)^2}) + \sum_i (A_i) \times (Y^i),$$

where,

[0066] X is the relative distance between a point on the aspheric surface spaced at a distance Y from an optical axis and the tangential plane at the aspheric surface vertex on the optical axis;

[0067] Y is the vertical distance from the point on the aspheric surface to the optical axis;

[0068] R is the curvature radius;

[0069] k is the conic coefficient; and

[0070] A_i is the i-th aspheric coefficient, and in the embodiments, i may be, but is not limited to, 4, 6, 8, 10, 12, 14 and 16.

[0071] In the photographing optical lens assembly of the image capturing unit according to the 1st embodiment, when a focal length of the photographing optical lens assembly is f,

an f-number of the photographing optical lens assembly is F_{no} , and half of a maximal field of view of the photographing optical lens assembly is $HFOV$, these parameters have the following values: $f=3.43$ mm; $F_{no}=1.49$; and $HFOV=39.5$ degrees.

[0072] When an Abbe number of the second lens element **120** is V_2 , the Abbe number of the third lens element **130** is V_3 , an Abbe number of the fourth lens element **140** is V_4 , the following condition is satisfied: $(V_2+V_4)/V_3=0.84$.

[0073] When a curvature radius of the object-side surface **141** of the fourth lens element **140** is R_7 , a curvature radius of the image-side surface **142** of the fourth lens element **140** is R_8 , the following condition is satisfied: $|(R_7-R_8)/(R_7+R_8)|=0.36$.

[0074] When a curvature radius of the object-side surface **151** of the fifth lens element **150** is R_9 , a curvature radius of the image-side surface **152** of the fifth lens element **150** is R_{10} , the following condition is satisfied: $(R_9+R_{10})/(R_9-R_{10})=-1.67$.

[0075] When a curvature radius of the object-side surface **161** of the sixth lens element **160** is R_{11} , a curvature radius of the image-side surface **162** of the sixth lens element **160** is R_{12} , the following condition is satisfied: $(R_{11}+R_{12})/(R_{11}-R_{12})=7.05$.

[0076] When a focal length of the second lens element **120** is f_2 , a focal length of the third lens element **130** is f_3 , the following condition is satisfied: $f_3/f_2=-0.69$.

[0077] When an entrance pupil diameter of the photographing optical lens assembly is EPD , a sum of the axial distances between any two adjacent lens elements is ΣAT , the following condition is satisfied: $EPD/\Sigma AT=3.04$.

[0078] When the entrance pupil diameter of the photographing optical lens assembly is EPD , a maximum image height of the photographing optical lens assembly is $ImgH$, the following condition is satisfied: $EPD/ImgH=0.79$.

[0079] When an axial distance between the object-side surface **111** of the first lens element **110** and the image surface **180** is TL , the maximum image height of the photographing optical lens assembly is $ImgH$, the following condition is satisfied: $TL/ImgH=1.67$.

[0080] When the axial distance between the object-side surface **111** of the first lens element **110** and the image surface **180** is TL , the following condition is satisfied: $TL=4.85$ mm.

[0081] The detailed optical data of the 1st embodiment are shown in Table 1 and the aspheric surface data are shown in Table 2 below.

TABLE 1

1st Embodiment							
$f = 3.43$ mm, $F_{no} = 1.49$, $HFOV = 39.5$ deg.							
Surface #		Curvature Radius	Thickness	Material	Index	Abbe #	Focal Length
0	Object	Plano	Infinity				
1	Ape. Stop	Plano	-0.379				
2	Lens 1	1.971 (ASP)	0.608	Plastic	1.544	55.9	3.97
3		20.123 (ASP)	0.072				
4	Lens 2	2.175 (ASP)	0.220	Plastic	1.639	23.5	-7.78
5		1.453 (ASP)	0.330				
6	Lens 3	5.257 (ASP)	0.605	Plastic	1.544	55.9	5.36
7		-6.297 (ASP)	0.235				
8	Lens 4	-1.849 (ASP)	0.369	Plastic	1.639	23.5	-5.88
9		-3.921 (ASP)	0.035				
10	Lens 5	2.764 (ASP)	0.767	Plastic	1.544	55.9	6.56
11		11.071 (ASP)	0.084				
12	Lens 6	1.380 (ASP)	0.515	Plastic	1.544	55.9	-16.31
13		1.038 (ASP)	0.500				
14	IR-cut filter	Plano	0.300	Glass	1.517	64.2	—
15		Plano	0.211				
16	Image	Plano	—				

Note:

Reference wavelength is 587.6 nm (d-line),

TABLE 2

Aspheric Coefficients						
Surface #	2	3	4	5	6	7
k=	1.4122E-01	5.6571E+01	-2.0854E+01	-1.0711E+01	-2.7135E+01	1.9460E+01
A4=	6.0019E-03	-1.3990E-01	-1.5961E-01	1.0006E-01	-2.1102E-02	-1.0022E-01
A6=	-2.0455E-02	3.8146E-01	3.9683E-01	-2.3491E-01	-2.0765E-02	1.0466E-01
A8=	5.2521E-02	-5.0071E-01	-5.2655E-01	4.8993E-01	2.4840E-02	-1.1443E-01
A10=	-6.5019E-02	3.7508E-01	3.7709E-01	-5.7385E-01	-5.0577E-02	-1.1064E-01
A12=	4.4922E-02	-1.5550E-01	-1.5573E-01	3.2447E-01	3.1913E-02	3.1028E-01
A14=	-1.2911E-02	2.6323E-02	2.9340E-02	-6.7935E-02	-3.2508E-03	-2.0637E-01
A16=	—	—	—	—	—	4.4824E-02

TABLE 2-continued

Aspheric Coefficients						
Surface #	8	9	10	11	12	13
k=	5.9832E-02	-1.6319E+00	-4.1407E+00	-8.4420E+01	-1.9910E+00	-2.0240E+00
A4=	-4.2234E-02	-1.7273E-01	-6.2428E-02	6.6726E-03	-3.8651E-01	-2.7404E-01
A6=	2.5525E-01	2.7202E-01	4.2657E-02	7.6962E-03	1.8957E-01	1.6911E-01
A8=	-4.1883E-01	-2.9855E-01	-7.7588E-02	-2.6784E-02	-5.8319E-02	-7.5249E-02
A10=	4.0121E-01	2.1903E-01	5.0411E-02	1.1361E-02	1.3599E-02	2.2681E-02
A12=	-1.7178E-01	-8.0629E-02	-1.8616E-02	-1.7195E-03	-2.2647E-03	-4.1865E-03
A14=	2.4733E-02	1.1357E-02	2.9934E-03	8.0226E-05	2.3040E-04	4.1525E-04
A16=	—	—	—	—	-1.0791E-05	-1.6809E-05

[0082] In Table 1, the curvature radius, the thickness and the focal length are shown in millimeters (mm). Surface numbers 0-16 represent the surfaces sequentially arranged from the object-side to the image-side along the optical axis. In Table 2, k represents the conic coefficient of the equation of the aspheric surface profiles. A4-A16 represent the aspheric coefficients ranging from the 4th order to the 16th order. The tables presented below for each embodiment are the corresponding schematic parameter and aberration curves, and the definitions of the tables are the same as Table 1 and Table 2 of the 1st embodiment. Therefore, an explanation in this regard will not be provided again.

2nd Embodiment

[0083] FIG. 3 is a schematic view of an image capturing unit according to the 2nd embodiment of the present disclosure. FIG. 4 shows, in order from left to right, spherical aberration curves, astigmatic field curves and a distortion curve of the image capturing unit according to the 2nd embodiment. In FIG. 3, the image capturing unit includes the photographing optical lens assembly (its reference numeral is omitted) of the present disclosure and an image sensor 290. The photographing optical lens assembly includes, in order from an object side to an image side an aperture stop 200, a first lens element 210, a second lens element 220, a third lens element 230, a fourth lens element 240, a fifth lens element 250, a sixth lens element 260, an IR-cut filter 270 and an image surface 280, wherein the photographing optical lens assembly has a total of six lens elements (210-260) with refractive power.

[0084] The first lens element 210 with positive refractive power has an object-side surface 211 being convex in a paraxial region thereof and an image-side surface 212 being convex in a paraxial region thereof. The first lens element 210 is made of plastic material and has the object-side surface 211 and the image-side surface 212 being both aspheric.

[0085] The second lens element 220 with negative refractive power has an object-side surface 221 being convex in a paraxial region thereof and an image-side surface 222 being

concave in a paraxial region thereof. The second lens element 220 is made of plastic material and has the object-side surface 221 and the image-side surface 222 being both aspheric.

[0086] The third lens element 230 with positive refractive power has an object-side surface 231 being convex in a paraxial region thereof and an image-side surface 232 being convex in a paraxial region thereof. The third lens element 230 is made of plastic material and has the object-side surface 231 and the image-side surface 232 being both aspheric.

[0087] The fourth lens element 240 with negative refractive power has an object-side surface 241 being concave in a paraxial region thereof and an image-side surface 242 being convex in a paraxial region thereof. The fourth lens element 240 is made of plastic material and has the object-side surface 241 and the image-side surface 242 being both aspheric.

[0088] The fifth lens element 250 with positive refractive power has an object-side surface 251 being convex in a paraxial region thereof and an image-side surface 252 being convex in a paraxial region thereof. The object-side surface 251 has at least one inflection point. The fifth lens element 250 is made of plastic material and has the object-side surface 251 and the image-side surface 252 being both aspheric.

[0089] The sixth lens element 260 with negative refractive power has an object-side surface 261 being convex in a paraxial region thereof and an image-side surface 262 being concave in a paraxial region thereof. The object-side surface 261 has at least one concave shape in an off-axis region thereof. The image-side surface 262 has at least one inflection point. The sixth lens element 260 is made of plastic material and has the object-side surface 261 and the image-side surface 262 being both aspheric.

[0090] The IR-cut filter 270 is made of glass and located between the sixth lens element 260 and the image surface 280, and will not affect the focal length of the photographing optical lens assembly. The image sensor 290 is disposed on or near the image surface 280 of the photographing optical lens assembly.

[0091] The detailed optical data of the 2nd embodiment are shown in Table 3 and the aspheric surface data are shown in Table 4 below.

TABLE 3

2nd Embodiment								
f = 3.47 mm, Fno = 1.42, HFOV = 38.9 deg.								
Surface #		Curvature Radius	Thickness	Material	Index	Abbe #	Focal Length	
0	Object	Plano	Infinity					
1	Ape. Stop	Plano	-0.374					
2	Lens 1	2.070 (ASP)	0.685	Plastic	1.544	55.9	3.71	

TABLE 3-continued

2nd Embodiment f = 3.47 mm, Fno = 1.42, HFOV = 38.9 deg.								
Surface #		Curvature Radius	Thickness	Material	Index	Abbe #	Focal Length	
3		-68.863 (ASP)	0.070					
4	Lens 2	2.766 (ASP)	0.220	Plastic	1.650	21.4	-6.57	
5		1.626 (ASP)	0.314					
6	Lens 3	5.802 (ASP)	0.559	Plastic	1.544	55.9	6.13	
7		-7.584 (ASP)	0.306					
8	Lens 4	-1.748 (ASP)	0.236	Plastic	1.650	21.4	-12.32	
9		-2.356 (ASP)	0.052					
10	Lens 5	4.295 (ASP)	1.044	Plastic	1.544	55.9	4.90	
11		-6.443 (ASP)	0.269					
12	Lens 6	1.183 (ASP)	0.268	Plastic	1.535	55.7	-4.14	
13		0.710 (ASP)	0.450					
14	IR-cut filter	Plano	0.300	Glass	1.517	64.2	—	
15		Plano	0.117					
16	Image	Plano	—					

Note:

Reference wavelength is 587.6 nm (d-line).

TABLE 4

Aspheric Coefficients						
Surface #	2	3	4	5	6	7
k=	3.5005E-02	9.0000E+01	-3.9254E+01	-1.5159E+01	-4.3861E+01	1.7941E+01
A4=	1.4791E-03	-1.0843E-01	-1.6103E-01	8.7774E-02	-4.4301E-02	-1.3329E-01
A6=	-1.8918E-02	3.4609E-01	4.1878E-01	-2.4368E-01	-2.9701E-02	1.3335E-01
A8=	5.1192E-02	-4.8786E-01	-5.3253E-01	5.1365E-01	3.6391E-02	-1.2620E-01
A10=	-6.7590E-02	3.7995E-01	3.6906E-01	-5.7649E-01	-3.7884E-02	-1.1038E-01
A12=	4.5033E-02	-1.6223E-01	-1.5065E-01	3.0845E-01	3.8097E-02	3.1886E-01
A14=	-1.2495E-02	2.7741E-02	2.8162E-02	-6.1447E-02	-1.0543E-02	-2.0260E-01
A16=	—	—	—	5.9819E-07	—	4.1293E-02
Surface #	8	9	10	11	12	13
k=	-1.1722E-01	-6.6142E+00	1.3599E+00	-8.6708E+01	-7.2246E+00	-3.5041E+00
A4=	-5.2943E-02	-1.4201E-01	-5.4346E-02	-1.5355E-02	-3.8340E-01	-2.3343E-01
A6=	2.5223E-01	2.5904E-01	4.0917E-02	1.8405E-02	1.8322E-01	1.5952E-01
A8=	-4.0850E-01	-3.0724E-01	-7.3444E-02	-2.5171E-02	-5.8503E-02	-7.5133E-02
A10=	3.9810E-01	2.1911E-01	5.1690E-02	1.0534E-02	1.3756E-02	2.2770E-02
A12=	-1.7521E-01	-7.9275E-02	-1.8690E-02	-1.8652E-03	-2.2384E-03	-4.1841E-03
A14=	2.7620E-02	1.1880E-02	2.6405E-03	1.1725E-04	2.3262E-04	4.1466E-04
A16=	1.6667E-06	-1.0252E-04	—	—	-1.1908E-05	-1.6812E-05

[0092] In the 2nd embodiment, the equation of the aspheric surface profiles of the aforementioned lens elements is the same as the equation of the 1st embodiment. Also, the definitions of these parameters shown in the following table are the same as those stated in the 1st embodiment with corresponding values for the 2nd embodiment, so an explanation in this regard will not be provided again.

[0093] Moreover, these parameters can be calculated from Table 3 and Table 4 as the following values and satisfy the following conditions:

2nd Embodiment		
f [mm]	3.47 (R11 + R12)/(R11 - R12)	4.00
Fno	1.42 f3/f2	-0.93
HFOV [deg.]	38.9 EPD/ΣAT	2.42
(V2 + V4)/V3	0.77 EPD/ImgH	0.86
(R7 - R8)/(R7 + R8)	0.15 TL/ImgH	1.72
(R9 + R10)/(R9 - R10)	-0.20 TL [mm]	4.89

3rd Embodiment

[0094] FIG. 5 is a schematic view of an image capturing unit according to the 3rd embodiment of the present disclosure. FIG. 6 shows, in order from left to right, spherical aberration curves, astigmatic field curves and a distortion curve of the image capturing unit according to the 3rd embodiment. In FIG. 5, the image capturing unit includes the photographing optical lens assembly (its reference numeral is omitted) of the present disclosure and an image sensor 390. The photographing optical lens assembly includes, in order from an object side to an image side, an aperture stop 300, a first lens element 310, a second lens element 320, a third lens element 330, a fourth lens element 340, a fifth lens element 350, a sixth lens element 360, a stop 301, an IR-cut filter 370 and an image surface 380, wherein the photographing optical lens assembly has a total of six lens elements (310-360) with refractive power.

[0095] The first lens element 310 with positive refractive power has an object-side surface 311 being convex in a

paraxial region thereof and an image-side surface **312** being concave in a paraxial region thereof. The first lens element **310** is made of plastic material and has the object-side surface **311** and the image-side surface **312** being both aspheric.

[0096] The second lens element **320** with negative refractive power has an object-side surface **321** being convex in a paraxial region thereof and an image-side surface **322** being concave in a paraxial region thereof. The second lens element **320** is made of plastic material and has the object-side surface **321** and the image-side surface **322** being both aspheric.

[0097] The third lens element **330** with positive refractive power has an object-side surface **331** being convex in a paraxial region thereof and an image-side surface **332** being convex in a paraxial region thereof. The third lens element **330** is made of plastic material and has the object-side surface **331** and the image-side surface **332** being both aspheric.

[0098] The fourth lens element **340** with negative refractive power has an object-side surface **341** being concave in a paraxial region thereof and an image-side surface **342** being convex in a paraxial region thereof. The fourth lens element **340** is made of plastic material and has the object-side surface **341** and the image-side surface **342** being both aspheric.

[0099] The fifth lens element **350** with positive refractive power has an object-side surface **351** being convex in a

paraxial region thereof and an image-side surface **352** being concave in a paraxial region thereof. The object-side surface **351** has at least one inflection point. The fifth lens element **350** is made of plastic material and has the object-side surface **351** and the image-side surface **352** being both aspheric.

[0100] The sixth lens element **360** with negative refractive power has an object-side surface **361** being convex in a paraxial region thereof and an image-side surface **362** being concave in a paraxial region thereof. The image-side surface **362** has at least one convex shape in an off-axis region thereof. The object-side surface **361** has at least one concave shape in an off-axis region thereof. The image-side surface **362** has at least one inflection point. The sixth lens element **360** is made of plastic material and has the object-side surface **361** and the image-side surface **362** being both aspheric.

[0101] The IR-cut filter **370** is made of glass and located between the sixth lens element **360** and the image surface **380**, and will not affect the focal length of the photographing optical lens assembly. The image sensor **390** is disposed on or near the image surface **380** of the photographing optical lens assembly.

[0102] The detailed optical data of the 3rd embodiment are shown in Table 5 and the aspheric surface data are shown in Table 6 below.

TABLE 5

3rd Embodiment								
f = 4.16 mm, Fno = 1.60, HFOV = 36.0 deg.								
Surface #		Curvature Radius	Thickness	Material	Index	Abbe #	Focal Length	
0	Object	Plano	Infinity					
1	Ape. Stop	Plano	-0.438					
2	Lens 1	1.814 (ASP)	0.814	Plastic	1.544	55.9	3.57	
3		23.091 (ASP)	0.078					
4	Lens 2	2.920 (ASP)	0.250	Plastic	1.639	23.5	-6.48	
5		1.655 (ASP)	0.388					
6	Lens 3	5.341 (ASP)	0.525	Plastic	1.544	55.9	6.76	
7		-11.392 (ASP)	0.167					
8	Lens 4	-2.097 (ASP)	0.512	Plastic	1.639	23.5	-8.58	
9		-3.719 (ASP)	0.051					
10	Lens 5	2.407 (ASP)	0.385	Plastic	1.544	55.9	11.80	
11		3.634 (ASP)	0.260					
12	Lens 6	2.259 (ASP)	0.607	Plastic	1.544	55.9	-11.21	
13		1.493 (ASP)	-0.280					
14	Stop	Plano	0.750					
15	IR-cut filter	Plano	0.300	Glass	1.517	64.2	—	
16		Plano	0.383					
17	Image	Plano	—					

Note:

Reference wavelength is 587.6 nm (d-line).

Effective radius of the stop (surface 14) is 2.55 mm.

TABLE 6

Aspheric Coefficients							
Surface #	2	3	4	5	6	7	
k=	4.2637E-02	7.9245E+01	-4.2972E+01	-1.5647E+01	-6.6411E+01	4.6427E+01	
A4=	2.0017E-03	-1.5911E-01	-1.6800E-01	1.2243E-01	-1.7453E-02	-9.1934E-02	
A6=	-2.1578E-02	3.8136E-01	4.0431E-01	-2.3634E-01	-3.9051E-02	8.8077E-02	
A8=	5.2613E-02	-4.9992E-01	-5.1115E-01	4.8869E-01	1.7449E-02	-1.1089E-01	
A10=	-6.8428E-02	3.7404E-01	3.8720E-01	-5.6289E-01	-4.7050E-02	-1.1100E-01	
A12=	4.2325E-02	-1.5335E-01	-1.6323E-01	3.4401E-01	3.3188E-02	3.0609E-01	
A14=	-1.0755E-02	2.5427E-02	2.8607E-02	-8.1664E-02	-2.7359E-03	-2.0898E-01	
A16=	—	—	—	—	—	4.5618E-02	

TABLE 6-continued

Aspheric Coefficients						
Surface #	8	9	10	11	12	13
k=	-1.1607E-01	-4.0097E+00	-9.2468E+00	-1.0204E+01	-1.0396E+00	-1.2010E+00
A4=	-4.0428E-02	-1.7733E-01	-4.1967E-02	-1.3340E-02	-3.7196E-01	-2.9335E-01
A6=	2.3682E-01	2.7450E-01	4.3614E-02	6.9513E-03	1.9124E-01	1.6884E-01
A8=	-4.2494E-01	-2.9924E-01	-9.0825E-02	-2.6266E-02	-5.8880E-02	-7.5273E-02
A10=	4.0657E-01	2.1719E-01	5.2345E-02	1.1498E-02	1.3483E-02	2.2643E-02
A12=	-1.6986E-01	-8.1239E-02	-1.6567E-02	-1.6960E-03	-2.2583E-03	-4.1698E-03
A14=	1.9300E-02	1.1645E-02	2.3787E-03	6.9594E-05	2.3268E-04	4.1779E-04
A16=	—	—	—	—	-1.0543E-05	-1.7262E-05

[0103] In the 3rd embodiment, the equation of the aspheric surface profiles of the aforementioned lens elements is the same as the equation of the 1st embodiment. Also, the definitions of these parameters shown in the following table are the same as those stated in the 1st embodiment with corresponding values for the 3rd embodiment, so an explanation in this regard will not be provided again.

[0104] Moreover, these parameters can be calculated from Table 5 and Table 6 as the following values and satisfy the following conditions:

3rd Embodiment		
f [mm]	4.16 (R11 + R12)/(R11 - R12)	4.89
Fno	1.60 f3/f2	-1.04
HFOV [deg.]	36.0 EPD/ΣAT	2.75
(V2 + V4)/V3	0.84 EPD/ImgH	0.84
(R7 - R8)/(R7 + R8)	0.28 TL/ImgH	1.69
(R9 + R10)/(R9 - R10)	-4.93 TL [mm]	5.19

4th Embodiment

[0105] FIG. 7 is a schematic view of an image capturing unit according to the 4th embodiment of the present disclosure. FIG. 8 shows, in order from left to right, spherical aberration curves, astigmatic field curves and a distortion curve of the image capturing unit according to the 4th embodiment. In FIG. 7, the image capturing unit includes the photographing optical lens assembly (its reference numeral is omitted) of the present disclosure and an image sensor 490. The photographing optical lens assembly includes, in order from an object side to an image side, a first lens element 410, an aperture stop 400, a second lens element 420, a third lens element 430, a fourth lens element 440, a fifth lens element 450, a sixth lens element 460, an IR-cut filter 470 and an image surface 480, wherein the photographing optical lens assembly has a total of six lens elements (410-460) with refractive power.

[0106] The first lens element 410 with positive refractive power has an object-side surface 411 being convex in a paraxial region thereof and an image-side surface 412 being concave in a paraxial region thereof. The first lens element 410 is made of plastic material and has the object-side surface 411 and the image-side surface 412 being both aspheric.

[0107] The second lens element 420 with negative refractive power has an object-side surface 421 being convex in a paraxial region thereof and an image-side surface 422 being concave in a paraxial region thereof. The second lens element 420 is made of plastic material and has the object-side surface 421 and the image-side surface 422 being both aspheric.

[0108] The third lens element 430 with positive refractive power has an object-side surface 431 being convex in a paraxial region thereof and an image-side surface 432 being concave in a paraxial region thereof. The third lens element 430 is made of plastic material and has the object-side surface 431 and the image-side surface 432 being both aspheric.

[0109] The fourth lens element 440 with negative refractive power has an object-side surface 441 being convex in a paraxial region thereof and an image-side surface 442 being concave in a paraxial region thereof. The fourth lens element 440 is made of plastic material and has the object-side surface 441 and the image-side surface 442 being both aspheric.

[0110] The fifth lens element 450 with positive refractive power has an object-side surface 451 being convex in a paraxial region thereof and an image-side surface 452 being concave in a paraxial region thereof. The object-side surface 451 has at least one inflection point. The fifth lens element 450 is made of plastic material and has the object-side surface 451 and the image-side surface 452 being both aspheric.

[0111] The sixth lens element 460 with negative refractive power has an object-side surface 461 being convex in a paraxial region thereof and an image-side surface 462 being concave in a paraxial region thereof. The object-side surface 461 has at least one concave shape in an off-axis region thereof. The image-side surface 462 has at least one inflection point. The sixth lens element 460 is made of plastic material and has the object-side surface 461 and the image-side surface 462 being both aspheric.

[0112] The IR-cut filter 470 is made of glass and located between the sixth lens element 460 and the image surface 480, and will not affect the focal length of the photographing optical lens assembly. The image sensor 490 is disposed on or near the image surface 480 of the photographing optical lens assembly.

[0113] The detailed optical data of the 4th embodiment are shown in Table 7 and the aspheric surface data are shown in Table 8 below.

TABLE 7

4th Embodiment f = 4.10 mm, Fno = 1.70, HFOV = 35.3 deg.								
Surface #		Curvature Radius	Thickness	Material	Index	Abbe #	Focal Length	
0	Object	Plano	Infinity					
1	Lens 1	1.817 (ASP)	0.802	Plastic	1.544	55.9	3.52	
2		29.842 (ASP)	0.035					
3	Ape. Stop	Plano	0.027					
4	Lens 2	3.072 (ASP)	0.304	Plastic	1.639	23.5	-6.18	
5		1.662 (ASP)	0.288					
6	Lens 3	6.373 (ASP)	0.658	Plastic	1.544	55.9	13.74	
7		41.578 (ASP)	0.070					
8	Lens 4	15.564 (ASP)	0.461	Plastic	1.639	23.5	-24.51	
9		7.716 (ASP)	0.273					
10	Lens 5	2.385 (ASP)	0.504	Plastic	1.544	55.9	9.67	
11		4.040 (ASP)	0.321					
12	Lens 6	3.639 (ASP)	0.534	Plastic	1.544	55.9	-5.93	
13		1.622 (ASP)	0.350					
14	IR-cut filter	Plano	0.300	Glass	1.517	64.2	—	
15		Plano	0.189					
16	Image	Plano						

Note:

Reference wavelength is 587.6 nm (d-line).

TABLE 8

Aspheric Coefficients						
Surface #	1	2	4	5	6	7
k=	2.0413E-02	8.5838E+00	-4.0271E+01	-1.2872E+01	-3.7622E+01	-9.0000E+01
A4=	1.9800E-03	-1.5474E-01	-1.6443E-01	1.1666E-01	-1.3413E-02	-7.4026E-02
A6=	-2.1824E-02	3.7877E-01	3.9385E-01	-2.3851E-01	-2.2469E-02	6.7851E-02
A8=	5.5066E-02	-4.9756E-01	-5.1203E-01	4.7733E-01	2.6695E-02	-1.0023E-01
A10=	-6.8455E-02	3.7956E-01	3.9244E-01	-5.5333E-01	-4.7173E-02	-1.0844E-01
A12=	3.9819E-02	-1.5791E-01	-1.5769E-01	3.4824E-01	3.1040E-02	3.0210E-01
A14=	-9.2346E-03	2.6861E-02	2.4416E-02	-8.5574E-02	-2.5096E-04	-2.1076E-01
A16=	—	—	—	—	—	4.8768E-02
Surface #	8	9	10	11	12	13
k=	-9.0000E+01	-6.3342E+00	-1.8395E+01	-8.6485E+01	-5.6459E-01	-1.0924E+00
A4=	-1.3987E-01	-2.0514E-01	-3.6915E-02	-1.3136E-02	-3.6519E-01	-2.9045E-01
A6=	2.3682E-01	2.6373E-01	3.1147E-02	1.3736E-02	1.9177E-01	1.6640E-01
A8=	-4.3391E-01	-2.9979E-01	-8.2651E-02	-2.6701E-02	-5.8936E-02	-7.4927E-02
A10=	4.0368E-01	2.1865E-01	5.1503E-02	1.1110E-02	1.3467E-02	2.2649E-02
A12=	-1.6690E-01	-8.0623E-02	-1.7897E-02	-1.7318E-03	-2.2598E-03	-4.1739E-03
A14=	2.0973E-02	1.1469E-02	2.7803E-03	8.6846E-05	2.3247E-04	4.1740E-04
A16=	—	—	—	—	-1.0458E-05	-1.7201E-05

[0114] In the 4th embodiment, the equation of the aspheric surface profiles of the aforementioned lens elements is the same as the equation of the 1st embodiment. Also, the definitions of these parameters shown in the following table are the same as those stated in the 1st embodiment with corresponding values for the 4th embodiment, so an explanation in this regard will not be provided again.

[0115] Moreover, these parameters can be calculated from Table 7 and Table 8 as the following values and satisfy the following conditions:

4th Embodiment		
f [mm]	4.10 (R11 + R12)/(R11 - R12)	2.61
Fno	1.70 f3/f2	-2.22
HFOV [deg]	35.3 EPD/ΣAT	2.38
(V2 + V4)/V3	0.84 EPD/ImgH	0.80

-continued

4th Embodiment		
(R7 - R8)/(R7 + R8)	0.34 TL/ImgH	1.71
(R9 + R10)/(R9 - R10)	-3.88 TL [mm]	5.12

5th Embodiment

[0116] FIG. 9 is a schematic view of an image capturing unit according to the 5th embodiment of the present disclosure. FIG. 10 shows, in order from left to right, spherical aberration curves, astigmatic field curves and a distortion curve of the image capturing unit according to the 5th embodiment. In FIG. 9, the image capturing unit includes the photographing optical lens assembly (its reference numeral is omitted) of the present disclosure and an image sensor 590.

The photographing optical lens assembly includes, in order from an object side to an image side, a first lens element **510**, an aperture stop **500**, a second lens element **520**, a third lens element **530**, a fourth lens element **540**, a fifth lens element **550**, a sixth lens element **560**, an IR-cut filter **570** and an image surface **580**, wherein the photographing optical lens assembly has a total of six lens elements (**510-560**) with refractive power.

[0117] The first lens element **510** with positive refractive power has an object-side surface **511** being convex in a paraxial region thereof and an image-side surface **512** being concave in a paraxial region thereof. The first lens element **510** is made of glass material and has the object-side surface **511** and the image-side surface **512** being both aspheric.

[0118] The second lens element **520** with negative refractive power has an object-side surface **521** being convex in a paraxial region thereof and an image-side surface **522** being concave in a paraxial region thereof. The second lens element **520** is made of plastic material and has the object-side surface **521** and the image-side surface **522** being both aspheric.

[0119] The third lens element **530** with positive refractive power has an object-side surface **531** being convex in a paraxial region thereof and an image-side surface **532** being convex in a paraxial region thereof. The third lens element **530** is made of plastic material and has the object-side surface **531** and the image-side surface **532** being both aspheric.

[0120] The fourth lens element **540** with negative refractive power has an object-side surface **541** being concave in a

paraxial region thereof and an image-side surface **542** being convex in a paraxial region thereof. The fourth lens element **540** is made of plastic material and has the object-side surface **541** and the image-side surface **542** being both aspheric.

[0121] The fifth lens element **550** with positive refractive power has an object-side surface **551** being convex in a paraxial region thereof and an image-side surface **552** being concave in a paraxial region thereof. The object-side surface **551** has at least one inflection point. The fifth lens element **550** is made of plastic material and has the object-side surface **551** and the image-side surface **552** being both aspheric.

[0122] The sixth lens element **560** with negative refractive power has an object-side surface **561** being convex in a paraxial region thereof and an image-side surface **562** being concave in a paraxial region thereof. The object-side surface **561** has at least one concave shape in an off-axis region thereof. The image-side surface **562** has at least one inflection point. The sixth lens element **560** is made of plastic material and has the object-side surface **561** and the image-side surface **562** being both aspheric.

[0123] The IR-cut filter **570** is made of glass and located between the sixth lens element **560** and the image surface **580**, and will not affect the focal length of the photographing optical lens assembly. The image sensor **590** is disposed on or near the image surface **580** of the photographing optical lens assembly.

[0124] The detailed optical data of the 5th embodiment are shown in Table 9 and the aspheric surface data are shown in Table 10 below.

TABLE 9

5th Embodiment							
f = 4.15 mm, Fno = 1.75, HFOV = 34.1 deg.							
Surface #		Curvature Radius	Thickness	Material	Index	Abbe #	Focal Length
0	Object	Plano	Infinity				
1	Lens 1	1.853 (ASP)	0.794	Glass	1.548	45.7	3.49
2		48.904 (ASP)	0.035				
3	Ape. Stop	Plano	0.026				
4	Lens 2	4.446 (ASP)	0.411	Plastic	1.655	20.5	-5.51
6		1.919 (ASP)	0.290				
7	Lens 3	5.011 (ASP)	0.592	Plastic	1.544	55.9	7.99
8		-31.581 (ASP)	0.120				
9	Lens 4	-5.093 (ASP)	0.508	Plastic	1.650	21.4	-12.30
10		-14.591 (ASP)	0.205				
11	Lens 5	2.383 (ASP)	0.520	Plastic	1.544	55.9	8.91
12		4.324 (ASP)	0.290				
13	Lens 6	3.193 (ASP)	0.548	Plastic	1.535	55.7	-6.33
14		1.545 (ASP)	0.350				
15	IR-cut filter	Plano	0.300	Glass	1.517	64.2	—
16		Plano	0.277				
17	Image	Plano	—				

Note:

Reference wavelength is 587.6 nm (d-line).

TABLE 10

Aspheric Coefficients							
Surface #	1	2	4	5	6	7	
k=	3.6709E-02	5.6275E+01	-7.4180E+01	-1.5680E+01	-1.2915E+01	-9.0000E+01	
A4=	4.9592E-04	-1.5517E-01	-1.5325E-01	1.2089E-01	-5.3974E-02	-1.0819E-01	
A6=	-2.0383E-02	3.8261E-01	3.9234E-01	-2.3070E-01	1.9197E-04	8.1949E-02	
A8=	5.4658E-02	-4.9667E-01	-5.1499E-01	4.7558E-01	2.1315E-02	-9.8815E-02	

TABLE 10-continued

Aspheric Coefficients						
A10=	-6.8633E-02	3.7908E-01	3.9838E-01	-5.6367E-01	-5.2966E-02	-1.0925E-01
A12=	4.0279E-02	-1.5924E-01	-1.6570E-01	3.5455E-01	2.8931E-02	3.0349E-01
A14=	-9.2732E-03	2.7524E-02	2.7319E-02	-8.6651E-02	2.0039E-03	-2.1027E-01
A16=	—	—	—	—	—	4.6029E-02
Surface #	8	9	10	11	12	13
k=	1.5604E+01	9.0000E+01	-2.1259E+01	-9.0000E+01	-5.3649E-01	-9.6175E-01
A4=	-1.1968E-01	-2.0387E-01	-3.3347E-02	-6.5424E-03	-3.6813E-01	-2.9636E-01
A6=	2.3974E-01	2.6943E-01	2.9379E-02	1.1685E-02	1.9200E-01	1.6725E-01
A8=	-4.2220E-01	-3.0023E-01	-8.1346E-02	-2.6630E-02	-5.8896E-02	-7.5017E-02
A10=	4.1056E-01	2.1808E-01	5.1511E-02	1.1174E-02	1.3471E-02	2.2656E-02
A12=	-1.6772E-01	-8.0688E-02	-1.7969E-02	-1.7283E-03	-2.2598E-03	-4.1723E-03
A14=	1.8129E-02	1.1617E-02	2.7206E-03	8.5036E-05	2.3243E-04	4.1748E-04
A16=	—	—	—	—	-1.0468E-05	-1.7215E-05

[0125] In the 5th embodiment, the equation of the aspheric surface profiles of the aforementioned lens elements is the same as the equation of the 1st embodiment. Also, the definitions of these parameters shown in the following table are the same as those stated in the 1st embodiment with corresponding values for the 5th embodiment, so an explanation in this regard will not be provided again.

[0126] Moreover, these parameters can be calculated from Table 9 and Table 10 as the following values and satisfy the following conditions:

5th Embodiment		
f [mm]	4.15 (R11 + R12)/(R11 - R12)	2.88
Fno	1.75 f3/f2	-1.45
HFOV [deg.]	34.1 EPD/ΣAT	2.45
(V2 + V4)/V3	0.75 EPD/ImgH	0.82
(R7 - R8)/(R7 + R8)	0.48 TL/ImgH	1.82
(R9 + R10)/(R9 - R10)	-3.45 TL [mm]	5.27

6th Embodiment

[0127] FIG. 11 is a schematic view of an image capturing unit according to the 6th embodiment of the present disclosure. FIG. 12 shows, in order from left to right, spherical aberration curves, astigmatic field curves and a distortion curve of the image capturing unit according to the 6th embodiment. In FIG. 11, the image capturing unit includes the photographing optical lens assembly (its reference numeral is omitted) of the present disclosure and an image sensor 690. The photographing optical lens assembly includes, in order from an object side to an image side, an aperture stop 600, a first lens element 610, a second lens element 620, a third lens element 630, a fourth lens element 640, a fifth lens element 650, a sixth lens element 660, an IR-cut filter 670 and an image surface 680, wherein the photographing optical lens assembly has a total of six lens elements (610-660) with refractive power.

[0128] The first lens element 610 with positive refractive power has an object-side surface 611 being convex in a paraxial region thereof and an image-side surface 612 being convex in a paraxial region thereof. The first lens element 610 is made of plastic material and has the object-side surface 611 and the image-side surface 612 being both aspheric.

[0129] The second lens element 620 with negative refractive power has an object-side surface 621 being convex in a paraxial region thereof and an image-side surface 622 being concave in a paraxial region thereof. The second lens element 620 is made of plastic material and has the object-side surface 621 and the image-side surface 622 being both aspheric.

[0130] The third lens element 630 with positive refractive power has an object-side surface 631 being convex in a paraxial region thereof and an image-side surface 632 being concave in a paraxial region thereof. The third lens element 630 is made of plastic material and has the object-side surface 631 and the image-side surface 632 being both aspheric.

[0131] The fourth lens element 640 with positive refractive power has an object-side surface 641 being concave in a paraxial region thereof and an image-side surface 642 being convex in a paraxial region thereof. The fourth lens element 640 is made of plastic material and has the object-side surface 641 and the image-side surface 642 being both aspheric.

[0132] The fifth lens element 650 with positive refractive power has an object-side surface 651 being convex in a paraxial region thereof and an image-side surface 652 being convex in a paraxial region thereof. The object-side surface 651 has at least one inflection point. The fifth lens element 650 is made of plastic material and has the object-side surface 651 and the image-side surface 652 being both aspheric.

[0133] The sixth lens element 660 with negative refractive power has an object-side surface 661 being convex in a paraxial region thereof and an image-side surface 662 being concave in a paraxial region thereof. The object-side surface 661 has at least one concave shape in an off-axis region thereof. The image-side surface 662 has at least one inflection point. The sixth lens element 660 is made of plastic material and has the object-side surface 661 and the image-side surface 662 being both aspheric.

[0134] The IR-cut filter 670 is made of glass and located between the sixth lens element 660 and the image surface 680, and will not affect the focal length of the photographing optical lens assembly. The image sensor 690 is disposed on or near the image surface 680 of the photographing optical lens assembly.

[0135] The detailed optical data of the 6th embodiment are shown in Table 11 and the aspheric surface data are shown in Table 12 below.

TABLE 11

6th Embodiment f = 3.56 mm, Fno = 1.50, HFOV = 37.2 deg.								
Surface #		Curvature Radius	Thickness	Material	Index	Abbe #	Focal Length	
0	Object	Plano	Infinity					
1	Ape. Stop	Plano	-0.348					
2	Lens 1	2.026 (ASP)	0.668	Plastic	1.544	55.9	3.70	
3		-313.924 (ASP)	0.085					
4	Lens 2	2.449 (ASP)	0.220	Plastic	1.650	21.4	-6.84	
5		1.523 (ASP)	0.335					
6	Lens 3	4.804 (ASP)	0.735	Plastic	1.544	55.9	9.83	
7		44.763 (ASP)	0.200					
8	Lens 4	-2.853 (ASP)	0.300	Plastic	1.650	21.4	47.91	
9		-2.722 (ASP)	0.030					
10	Lens 5	4.800 (ASP)	0.820	Plastic	1.530	55.8	7.59	
11		-23.385 (ASP)	0.284					
12	Lens 6	1.895 (ASP)	0.373	Plastic	1.535	55.7	-4.53	
13		0.990 (ASP)	0.400					
14	IR-cut filter	Plano	0.300	Glass	1.517	64.2	—	
15		Plano	0.214					
16	Image	Plano	—					

Note:

Reference wavelength is 587.6 nm (d-line).

TABLE 12

Aspheric Coefficients						
Surface #	2	3	4	5	6	7
k=	-2.5231E-02	-9.0000E+01	-3.4012E+01	-1.4671E+01	-8.8482E+01	9.0000E+01
A4=	-2.1697E-03	-1.2879E-01	-1.5996E-01	1.0139E-01	-1.2058E-02	-1.6019E-01
A6=	-1.5564E-02	3.6044E-01	4.2046E-01	-2.4722E-01	-3.2009E-02	1.3916E-01
A8=	4.8537E-02	-4.9037E-01	-5.2503E-01	5.1516E-01	3.3267E-02	-1.2590E-01
A10=	-7.0191E-02	3.7513E-01	3.6553E-01	-5.7294E-01	-3.6492E-02	-1.1590E-01
A12=	4.9244E-02	-1.6061E-01	-1.5567E-01	3.0735E-01	3.8758E-02	3.1538E-01
A14=	-1.4568E-02	2.7724E-02	3.0825E-02	-6.2480E-02	-1.1150E-02	-2.0246E-01
A16=	—	—	—	-4.9113E-05	—	4.3038E-02
Surface #	8	9	10	11	12	13
k=	5.0059E-01	-2.1742E+01	4.8351E+00	-9.0000E+01	-6.1873E+00	-3.1781E+00
A4=	-6.0561E-02	-1.3974E-01	-2.7219E-02	-7.5015E-03	-3.5794E-01	-2.3224E-01
A6=	2.3450E-01	2.6777E-01	2.8366E-02	2.2581E-02	1.8315E-01	1.5911E-01
A8=	-4.1468E-01	-3.0719E-01	-7.9098E-02	-2.6483E-02	-5.8871E-02	-7.5174E-02
A10=	3.9908E-01	2.1794E-01	5.4081E-02	1.0544E-02	1.3707E-02	2.2763E-02
A12=	-1.7443E-01	-7.9814E-02	-1.7458E-02	-1.8774E-03	-2.2449E-03	-4.1782E-03
A14=	2.7579E-02	1.1768E-02	2.0646E-03	1.1865E-04	2.3290E-04	4.1536E-04
A16=	-1.6343E-04	-1.3763E-04	—	—	-1.1228E-05	-1.6933E-05

[0136] In the 6th embodiment, the equation of the aspheric surface profiles of the aforementioned lens elements is the same as the equation of the 1st embodiment. Also, the definitions of these parameters shown in the following table are the same as those stated in the 1st embodiment with corresponding values for the 6th embodiment, so an explanation in this regard will not be provided again.

[0137] Moreover, these parameters can be calculated from Table 11 and Table 12 as the following values and satisfy the following conditions:

6th Embodiment		
f [mm]	3.56 (R11 + R12)/(R11 - R12)	3.19
Fno	1.50 f3/f2	-1.44
HFOV [deg.]	37.2 EPD/ΣAT	2.54
(V2 + V4)/V3	0.77 EPD/ImgH	0.85

-continued

6th Embodiment		
(R7 - R8)/(R7 + R8)	0.02 TL/ImgH	1.77
(R9 + R10)/(R9 - R10)	-0.66 TL [mm]	4.96

7th Embodiment

[0138] FIG. 13 is a schematic view of an image capturing unit according to the 7th embodiment of the present disclosure. FIG. 14 shows, in order from left to right, spherical aberration curves, astigmatic field curves and a distortion curve of the image capturing unit according to the 7th embodiment. In FIG. 13, the image capturing unit includes the photographing optical lens assembly (its reference numeral is omitted) of the present disclosure and an image

sensor **790**. The photographing optical lens assembly includes, in order from an object side to an image side, an aperture stop **700**, a first lens element **710**, a second lens element **720**, a third lens element **730**, a fourth lens element **740**, a fifth lens element **750**, a sixth lens element **760**, a stop **701**, an IR-cut filter **770** and an image surface **780**, wherein the photographing optical lens assembly has a total of six lens elements (**710-760**) with refractive power.

[0139] The first lens element **710** with positive refractive power has an object-side surface **711** being convex in a paraxial region thereof and an image-side surface **712** being concave in a paraxial region thereof. The first lens element **710** is made of plastic material and has the object-side surface **711** and the image-side surface **712** being both aspheric.

[0140] The second lens element **720** with negative refractive power has an object-side surface **721** being convex in a paraxial region thereof and an image-side surface **722** being concave in a paraxial region thereof. The second lens element **720** is made of plastic material and has the object-side surface **721** and the image-side surface **722** being both aspheric.

[0141] The third lens element **730** with positive refractive power has an object-side surface **731** being convex in a paraxial region thereof and an image-side surface **732** being convex in a paraxial region thereof. The third lens element **730** is made of plastic material and has the object-side surface **731** and the image-side surface **732** being both aspheric.

[0142] The fourth lens element **740** with negative refractive power has an object-side surface **741** being concave in a

paraxial region thereof and an image-side surface **742** being convex in a paraxial region thereof. The fourth lens element **740** is made of plastic material and has the object-side surface **741** and the image-side surface **742** being both aspheric.

[0143] The fifth lens element **750** with positive refractive power has an object-side surface **751** being convex in a paraxial region thereof and an image-side surface **752** being concave in a paraxial region thereof. The object-side surface **751** has at least one inflection point. The fifth lens element **750** is made of plastic material and has the object-side surface **751** and the image-side surface **752** being both aspheric.

[0144] The sixth lens element **760** with negative refractive power has an object-side surface **761** being convex in a paraxial region thereof and an image-side surface **762** being concave in a paraxial region thereof. The object-side surface **761** has at least one concave shape in an off-axis region thereof. The image-side surface **762** has at least one inflection point. The sixth lens element **760** is made of plastic material and has the object-side surface **761** and the image-side surface **762** being both aspheric.

[0145] The IR-cut filter **770** is made of glass and located between the sixth lens element **760** and the image surface **780**, and will not affect the focal length of the photographing optical lens assembly. The image sensor **790** is disposed on or near the image surface **780** of the photographing optical lens assembly.

[0146] The detailed optical data of the 7th embodiment are shown in Table 13 and the aspheric surface data are shown in Table 14 below.

TABLE 13

7th Embodiment								
f = 4.17 mm, Fno = 1.80, HFOV = 36.0 deg.								
Surface #		Curvature Radius	Thickness	Material	Index	Abbe #	Focal Length	
0	Object	Plano	Infinity					
1	Ape. Stop	Plano	-0.410					
2	Lens 1	1.795 (ASP)	0.733	Plastic	1.544	55.9	3.57	
3		20.110 (ASP)	0.087					
4	Lens 2	2.956 (ASP)	0.250	Plastic	1.639	23.5	-6.49	
5		1.669 (ASP)	0.405					
6	Lens 3	5.382 (ASP)	0.539	Plastic	1.544	55.9	6.69	
7		-10.859 (ASP)	0.182					
8	Lens 4	-2.094 (ASP)	0.504	Plastic	1.639	23.5	-8.59	
9		-3.705 (ASP)	0.060					
10	Lens 5	2.364 (ASP)	0.386	Plastic	1.544	55.9	12.19	
11		3.462 (ASP)	0.268					
12	Lens 6	2.237 (ASP)	0.611	Plastic	1.544	55.9	-10.95	
13		1.470 (ASP)	-0.250					
14	Stop	Plano	0.750					
15	IR-cut filter	Plano	0.300	Glass	1.517	64.2	—	
16		Plano	0.353					
17	Image	Plano	—					

Note:

Reference wavelength is 587.6 nm (d-line).

Effective radius of the stop (surface 14) is 2.56 mm.

TABLE 14

Aspheric Coefficients						
Surface #	2	3	4	5	6	7
k=	3.7666E-02	7.6949E+01	-4.2606E+01	-1.5588E+01	-6.8478E+01	4.6871E+01
A4=	-2.8381E-03	-1.5721E-01	-1.6608E-01	1.1490E-01	-1.6509E-02	-7.6266E-02
A6=	3.6477E-03	3.9997E-01	4.1811E-01	-2.0761E-01	-5.8234E-02	-4.8222E-03
A8=	-9.8763E-04	-5.7563E-01	-5.7767E-01	4.2517E-01	6.0377E-02	8.8906E-02

TABLE 14-continued

Aspheric Coefficients						
A10=	-1.1899E-02	4.8679E-01	4.9992E-01	-4.8247E-01	-1.0158E-01	-3.2766E-01
A12=	1.4694E-02	-2.2892E-01	-2.4861E-01	2.9499E-01	7.6362E-02	4.3738E-01
A14=	-6.2560E-03	4.3869E-02	5.2410E-02	-7.0579E-02	-1.7124E-02	-2.5337E-01
A16=	—	—	—	—	—	5.2432E-02
Surface #	8	9	10	11	12	13
k=	-1.1789E-01	-4.2640E+00	-9.3413E+00	-9.9440E+00	-1.0562E+00	-1.2539E+00
A4=	-2.6867E-02	-1.7101E-01	-3.5494E-02	-4.8246E-03	-3.6435E-01	-2.9239E-01
A6=	1.6139E-01	2.4727E-01	2.8151E-02	-6.2018E-03	1.7797E-01	1.6716E-01
A8=	-2.6916E-01	-2.5469E-01	-7.8851E-02	-1.8615E-02	-5.0294E-02	-7.4263E-02
A10=	2.6023E-01	1.8406E-01	4.8697E-02	9.7038E-03	1.0717E-02	2.2440E-02
A12=	-1.0677E-01	-6.9722E-02	-1.6553E-02	-1.5638E-03	-1.7844E-03	-4.1697E-03
A14=	9.2343E-03	1.0126E-02	2.5015E-03	7.2802E-05	1.9184E-04	4.2226E-04
A16=	—	—	—	—	-9.1871E-06	-1.7638E-05

[0147] In the 7th embodiment, the equation of the aspheric surface profiles of the aforementioned lens elements is the same as the equation of the 1st embodiment. Also, the definitions of these parameters shown in the following table are the same as those stated in the 1st embodiment with corresponding values for the 7th embodiment, so an explanation in this regard will not be provided again.

[0148] Moreover, these parameters can be calculated from Table 13 and Table 14 as the following values and satisfy the following conditions:

7th Embodiment		
f [mm]	$4.17 (R11 + R12)/(R11 - R12)$	4.83
Fno	$1.80 f/f2$	-1.03
HFOV [deg.]	$36.0 EPD/\Sigma AT$	2.31
$(V2 + V4)/V3$	$0.84 EPD/ImgH$	0.75
$ (R7 - R8)/(R7 + R8) $	$0.28 TL/ImgH$	1.68
$(R9 + R10)/(R9 - R10)$	$-5.31 TL [mm]$	5.18

[0149] The foregoing image capturing unit is able to be installed in, but not limited to, a mobile device, including smart phones, personal tablet computers and wearable apparatus. According to the present disclosure, the photographing optical lens assembly comprising six lens elements can control the entrance pupil diameter thereof and the air gaps between two lens elements adjacent to each other, for arranging the lens elements tightly in a limited amount of space as well as increasing the amount of incoming light. When specific conditions are satisfied, the interior space of the photographing optical lens assembly can be utilized efficiently so that it is favorable for reducing the total track length thereof and providing sufficient amount of incident light, thereby increasing the relative illumination of the image.

[0150] The foregoing description, for the purpose of explanation, has been described with reference to specific embodiments. It is to be noted that TABLES 1-14 show different data of the different embodiments; however, the data of the different embodiments are obtained from experiments. The embodiments were chosen and described in order to best explain the principles of the disclosure and its practical applications, to thereby enable others skilled in the art to best utilize the disclosure and various embodiments with various modifications as are suited to the particular use contemplated. The embodiments depicted above and the appended drawings are exemplary and are not intended to be exhaustive or to limit

the scope of the present disclosure to the precise forms disclosed. Many modifications and variations are possible in view of the above teachings.

What is claimed is:

1. A photographing optical lens assembly comprising, in order from an object side to an image side:

- a first lens element having positive refractive power;
- a second lens element with negative refractive power having an object-side surface being convex in a paraxial region thereof and an image-side surface being concave in a paraxial region thereof;
- a third lens element having refractive power;
- a fourth lens element having refractive power;
- a fifth lens element with refractive power having an object-side surface being convex in a paraxial region thereof, wherein both of the object-side surface and an image-side surface of the fifth lens element are aspheric; and
- a sixth lens element with negative refractive power having an image-side surface being concave in a paraxial region thereof, wherein the image-side surface of the sixth lens element has at least one inflection point, and both of an object-side surface and the image-side surface of the sixth lens element are aspheric;

wherein the photographing optical lens assembly has a total of six lens elements with refractive power, an entrance pupil diameter of the photographing optical lens assembly is EPD, a sum of axial distances between any two lens elements adjacent to each other is ΣAT , a curvature radius of the object-side surface of the fifth lens element is R9, a curvature radius of the image-side surface of the fifth lens element is R10, a curvature radius of the object-side surface of the sixth lens element is R11, a curvature radius of the image-side surface of the sixth lens element is R12, and the following conditions are satisfied:

$$2.20 < EPD/\Sigma AT;$$

$$(R9+R10)/(R9-R10) < -0.15; \text{ and}$$

$$1.20 < (R11+R12)/(R11-R12).$$

2. The photographing optical lens assembly of claim 1, wherein the third lens element has positive refractive power, an f-number of the photographing optical lens assembly is Fno, and the following condition is satisfied:

$$Fno < 1.85.$$

3. The photographing optical lens assembly of claim 1, wherein the fifth lens element has positive refractive power, the curvature radius of the object-side surface of the fifth lens element is R9, the curvature radius of the image-side surface of the fifth lens element is R10, and the following condition is satisfied:

$$(R9+R10)/(R9-R10) \leq -1.00.$$

4. The photographing optical lens assembly of claim 1, wherein a focal length of the second lens element is f2, a focal length of the third lens element is f3, and the following condition is satisfied:

$$-1.50 < \beta/f2.$$

5. The photographing optical lens assembly of claim 1, wherein an Abbe number of the second lens element is V2, an Abbe number of the third lens element is V3, an Abbe number of the fourth lens element is V4, and the following condition is satisfied:

$$(V2+V4)/V3 < 1.00.$$

6. The photographing optical lens assembly of claim 1, wherein a curvature radius of an object-side surface of the fourth lens element is R7, a curvature radius of an image-side surface of the fourth lens element is R8, and the following condition is satisfied:

$$|(R7-R8)/(R7+R8)| < 0.50.$$

7. The photographing optical lens assembly of claim 1, wherein the entrance pupil diameter of the photographing optical lens assembly is EPD, a maximum image height of the photographing optical lens assembly is ImgH, and the following condition is satisfied:

$$0.65 < EPD/ImgH < 1.50.$$

8. An image capturing unit, comprising:

the photographing optical lens assembly of claim 1; and
an image sensor, wherein the image sensor is disposed on the image side of the photographing optical lens assembly.

9. A mobile device, comprising:

the image capturing unit of claim 8.

10. A photographing optical lens assembly comprising, in order from an object side to an image side:

- a first lens element having positive refractive power;
- a second lens element having negative refractive power;
- a third lens element having refractive power;
- a fourth lens element having refractive power;
- a fifth lens element having refractive power, wherein both of an object-side surface and an image-side surface of the fifth lens element are aspheric; and
- a sixth lens element with refractive power having an image-side surface being concave in a paraxial region thereof, wherein the image-side surface of the sixth lens element has at least one inflection point, and both of an object-side surface and the image-side surface of the sixth lens element are aspheric;

wherein the photographing optical lens assembly has a total of six lens elements with refractive power, an entrance pupil diameter of the photographing optical lens assembly is EPD, a sum of axial distances between any two lens elements adjacent to each other is ΣAT , a curvature radius of the object-side surface of the fifth

lens element is R9, a curvature radius of the image-side surface of the fifth lens element is R10, and the following conditions are satisfied:

$$2.40 < EPD/\Sigma AT; \text{ and}$$

$$(R9+R10)/(R9-R10) < -0.15.$$

11. The photographing optical lens assembly of claim 10, wherein the fourth lens element has negative refractive power, the entrance pupil diameter of the photographing optical lens assembly is EPD, a maximum image height of the photographing optical lens assembly is ImgH, and the following condition is satisfied:

$$0.65 < EPD/ImgH < 1.50.$$

12. The photographing optical lens assembly of claim 10, wherein the object-side surface of the fifth lens element has at least one inflection point.

13. The photographing optical lens assembly of claim 10, wherein the image-side surface of the fifth lens element is concave in a paraxial region thereof.

14. The photographing optical lens assembly of claim 10, wherein the third lens element has an object-side surface being convex in a paraxial region thereof, the fourth lens element has an object-side surface being concave in a paraxial region thereof and an image-side surface being convex in a paraxial region thereof.

15. The photographing optical lens assembly of claim 10, wherein there is an air gap between any two of the first lens element, the second lens element, the third lens element, the fourth lens element, the fifth lens element and the sixth lens element that are adjacent to each other in a paraxial region thereof.

16. The photographing optical lens assembly of claim 10, wherein an f-number of the photographing optical lens assembly is Fno, and the following condition is satisfied:

$$Fno < 1.65.$$

17. The photographing optical lens assembly of claim 10, wherein a focal length of the second lens element is f2, a focal length of the third lens element is f3, and the following condition is satisfied:

$$-1.50 < \beta/f2.$$

18. A photographing optical lens assembly comprising, in order from an object side to an image side:

- a first lens element with positive refractive power having an object-side surface being convex in a paraxial region thereof;
- a second lens element having refractive power;
- a third lens element having refractive power;
- a fourth lens element having refractive power;
- a fifth lens element having refractive power, wherein both of an object-side surface and an image-side surface of the fifth lens element are aspheric; and
- a sixth lens element with refractive power having an image-side surface being concave in a paraxial region thereof, wherein the image-side surface of the sixth lens element has at least one inflection point, and both of an object-side surface and the image-side surface of the sixth lens element are aspheric;

wherein the photographing optical lens assembly has a total of six lens elements with refractive power, an entrance pupil diameter of the photographing optical lens assembly is EPD, a sum of axial distances between

any two adjacent lens elements is ΣAT , an axial distance between the object-side surface of the first lens element and an image surface is TL , a maximum image height of the photographing optical lens assembly is $ImgH$, and the following conditions are satisfied:

$$2.20 < EPD/\Sigma AT;$$

$$0.73 < EPD/ImgH < 1.10; \text{ and}$$

$$TL/ImgH < 2.50.$$

19. The photographing optical lens assembly of claim **18**, wherein the second lens element has negative refractive power.

20. The photographing optical lens assembly of claim **18**, wherein the sixth lens element has negative refractive power.

21. The photographing optical lens assembly of claim **18**, wherein the object-side surface of the sixth lens element is convex in a paraxial region thereof, and the object-side surface of the sixth lens element has at least one concave shape in an off-axis region thereof.

22. The photographing optical lens assembly of claim **18**, wherein the entrance pupil diameter of the photographing optical lens assembly is EPD , the maximum image height of the photographing optical lens assembly is $ImgH$, and the following condition is satisfied:

$$0.78 < EPD/ImgH < 0.90.$$

23. The photographing optical lens assembly of claim **18**, wherein an f-number of the photographing optical lens assembly is Fno , and the following condition is satisfied:

$$Fno < 1.55.$$

24. The photographing optical lens assembly of claim **18**, wherein the axial distance between the object-side surface of the first lens element and the image surface is TL , and the following condition is satisfied:

$$TL < 6.5 \text{ mm.}$$

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